1. Loft Language Reference

A statically-typed scripting language with null safety and built-in parallel execution.

Version 0.8.3

Every code example in this document is an executable part of the test suite.

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2. Getting Started

Loft is a lightweight scripting language with null safety, built-in parallel execution, and a fast interpreter called **loft**. This page shows how to install loft and write your first program.

2.0.1. Prerequisites

Building from source requires the Rust toolchain (Rust 1.82 or later). Install it from rustup.rs if you do not already have it:

```
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
```

2.0.2. Install from source

The quickest way to get the `loft` binary on your PATH:

```
cargo install --git https://github.com/jjstwerff/loft --bin loft
```

This compiles loft in release mode and places the binary in `~/.cargo/bin/`, which is already on your PATH if you used rustup.

2.0.3. Build locally

Clone the repository and build with Cargo:

```
git clone https://github.com/jjstwerff/loft
cd loft
cargo build --release
```

The binary is at `target/release/loft`. Copy it anywhere on your PATH, for example:

```
cp target/release/loft ~/.local/bin/
```

2.0.4. Verify the installation

```
loft --version
```

2.0.5. Hello, World

Create a file called `hello.loft`:

```
fn main() {
    println("Hello, Loft!");
}
```

Run it:

```
loft hello.loft
```

```
Hello, Loft!
```

2.0.6. A slightly bigger program

Variables, loops, and string interpolation all work without any ceremony:

```
fn main() {
    // Variables are mutable by default; types are inferred.
    total = 0;
    for n in 1..=10 {
        total += n;
    }
    println("Sum 1..10 = {total}");

    // Vectors use square-bracket literals.
    words = ["loft", "is", "fast"];
    for w in words {
        print("{w} ");
    }
    println("");
}
```

2.0.7. Command-line flags

Run `loft --help` for the full list. The most commonly used flags are:

2.0.8. Running your program as a native binary

The interpreter runs programs immediately, but for compute-heavy work you can compile to a native binary instead. This takes a few seconds the first time (it calls `rustc` in the background) but runs 10–50× faster:

```
loft --native myprogram.loft
```

To see the generated Rust code without running it:

```
loft --native-emit myprogram.rs
cat myprogram.rs
```

Native compilation requires Rust 1.85 or later on your PATH. If `rustc` is not found, loft prints a clear error message telling you how to install it.

2.0.9. Standard library

The standard library is a set of `.loft` files bundled alongside the loft binary. They are loaded automatically before your program runs — you do not need any `import` or `use` statement to call `println`, `assert`, or any other built-in function.

The full function reference is in the Standard Library section of these docs.

2.0.10. Using libraries

Place a library file (e.g. `lib/myutil.loft`) next to your program, then bring it into scope:

```
use myutil;
```

```
fn main() {
    result = myutil::compute(42);
    println("{result}");
}
```

See Libraries for the full search path and naming rules.

2.0.11. Editor setup

Loft syntax highlighting extensions are planned. In the meantime:

- **VS Code** — use Rust syntax highlighting as a close approximation.
- **Vim / Neovim** — associate `*.loft` with Rust via `autocmd BufRead *.loft set filetype=rust` in your `vimrc`.
- **Any editor** — the Web IDE (in progress) provides syntax highlighting and navigation in the browser.

2.0.12. Next steps

- Read the language overview starting with Keywords and control flow.
- See vs Rust if you are coming from Rust.
- Browse the Standard Library reference for built-in functions.
- Check the Roadmap for planned features.

3. vs Rust

Loft is inspired by Rust but designed for general-purpose scripting with a shorter learning curve and less ceremony. It trades some of Rust's safety guarantees and expressive power for a simpler mental model. This page documents the most important differences so that Rust programmers know exactly what they gain and what they give up.

3.0.1. Variables — no let or mut

```
x = 42
x += 1
const y = 42   // opt-in: locked in debug builds
```

```
let mut x = 42;
x += 1;
let y = 42;     // immutable by default; compiler-enforced
```

Upside Less boilerplate. No need to decide upfront whether a variable will be mutated. Variables are mutable by default, so refactoring is frictionless. Use const to signal that a value should not change — in debug builds the runtime locks the store immediately after initialisation, catching accidental writes early.

Downside Immutability is opt-in, not the default. Rust makes variables immutable unless you write mut, catching accidents at compile time for free. Loft's const lock is only checked at runtime and only in debug builds, so it provides less of a safety net.

3.0.2. Null instead of Option<T>

```
struct Point {
    label: text,          // nullable
    x: integer not null,  // never null
}
p = Point { x: 1 };
assert(p.label == null, "absent");
```

```
struct Point {
    label: Option<String>,  // explicit absence
    x: i32,                 // always present
}
let p = Point { label: None, x: 1 };
assert!(p.label.is_none());
```

Upside Natural for database records where fields are often absent. No Some(x)/None wrapping. Conditionals on null are concise: if v == null.

Downside Null is opt-out, not opt-in. A field is nullable unless explicitly marked not null, so forgetting the annotation leaves a potential null silently. In Rust, Option is visible in the type and forces handling at every call site.

3.0.3. Parameter mutability — const and & instead of ownership

```
// &: vector growth (append) is visible to caller
fn push_two(v: &vector<integer>) {
    v += [1, 2];  // caller sees the change
}
// const: read-only (locked in debug builds)
fn count(v: const vector<integer>) - integer {
    length(v)
}
// &: explicit write-back for primitives
fn add_to(n: &integer, delta: integer) {
    n += delta;
}
```

```
fn push_two(v: &mut Vec<i32>) {
    v.push(1);
    v.push(2);
}
fn count(v: &Vec<i32>) -> usize {
    v.len()
}
fn add_to(n: &mut i32, delta: i32) {
    *n += delta;
}
```

Upside No borrow-checker errors. No lifetime annotations. No ownership transfer to reason about. Struct field mutations through a parameter are always visible to the caller. Use & on a collection parameter to allow vector growth (append) to propagate back. const parameters signal read-only intent, and in debug builds the runtime locks the store for the duration of the call — enough to catch most accidents during development.

Downside The compile-time guarantees are much weaker. Rust's borrow checker statically proves no aliased mutations, no dangling references, and no data races — before the program ever runs. Loft's const is a debug-only runtime check. Aliasing is unchecked at compile time, and the engine's Rust runtime is what truly enforces memory safety.

3.0.4. ^ is XOR; exponentiation uses pow()

```
area = PI * pow(r, 2.0)  // exponentiation via pow()
bits = a | b              // bitwise OR
mask = a & b              // bitwise AND
xor  = a ^ b              // bitwise XOR
```

```
let area = PI * r.powi(2); // or r.powf(2.0)
let bits = a | b;          // bitwise OR
let mask = a & b;          // bitwise AND
let xor  = a ^ b;          // bitwise XOR
```

Upside ^ behaves exactly as Rust developers expect — it is bitwise XOR. Bitwise operator precedence matches Rust and C: | (loose) → ^ → & → shifts → arithmetic (tight).

Downside Exponentiation requires a function call: pow(base, exp) for float/single — there is no ** or ^ exponentiation operator. Integer exponentiation is not in the standard library; compute it with a loop or cast to float first.

3.0.5. No loop keyword — use while or for + break

```
while !done() { step(); }  // while works

// Infinite loop workaround — use a long range:
for _ in 0l..9223372036854775807l {  // long: ~9.2×10^18
    if should_exit() { break; }
    step();
}
```

```
while !done() { step(); }

loop {           // truly infinite — no bound needed
    if should_exit() { break; }
    step();
}
```

Upside while condition { } works exactly as expected. Every for loop has an iteration variable, making it easy to add index tracking or a cycle limit without restructuring.

Downside There is no loop { } keyword for an unconditionally infinite loop. The workaround — a for loop over a long range — is verbose but practically unbounded: the 64-bit maximum (9.2×10^18) exceeds any realistic event-loop iteration count.

3.0.6. Filtered loops — for … if …

```
for x in items if x > 0 {
    total += x;
}
```

```
for x in items.iter().filter(|&&x| x > 0) {
    total += x;
}
```

Upside Reads like natural language. No closure syntax, no double-reference in the filter predicate. v#remove inside a filtered loop also safely removes the current element while iterating — something that requires careful index management in Rust.

Downside The inline if filter is limited to the loop it lives in. For multi-stage pipelines, loft now offers map(v, fn f), filter(v, fn pred), and reduce(v, fn f, init) as composable higher-order functions — see section 11. Rust's lazy iterator chain (.filter().map().take_while()) avoids intermediate allocations; loft's higher-order functions allocate a new vector at each stage.

3.0.7. Loop attributes: #first, #count, #index

```
for x in 1..=5 {
    if !x#first { b += ", "; }
    b += "{x#count}:{x}";
}
```

```
for (i, x) in (1..=5).enumerate() {
    if i != 0 { b.push_str(", "); }
    b.push_str(&format!("{i}:{x}"));
}
```

Upside No tuple destructuring needed. x#first reads as "is this the first x", which is self-explanatory. Eliminates helper counter variables for the common pattern of comma-separated output.

Downside The #attribute syntax is unique to loft and unfamiliar to everyone else. Rust's .enumerate() composes freely with other adapters; loft's attributes are only available on the loop variable of the innermost loop.

3.0.8. Named loop break — x#break

```
for x in 1..5 {
    for y in 1..5 {
        if x * y >= 6 { x#break; }
    }
}
```

```
'outer: for x in 1..5 {
    for y in 1..5 {
        if x * y >= 6 { break 'outer; }
    }
}
```

Upside Reuses the existing loop variable name as the label. No separate 'label declaration at the loop head. The name x#break reads as "break out of the loop over x".

Downside Rust's lifetime-label syntax 'outer is explicit and visually separate from the loop body. x#break is easy to confuse with the loop attributes x#first and x#count, which have completely different semantics.

3.0.9. Methods by self name, not impl blocks

```
fn greet(self: Person) - text {
    "Hello, {self.name}!"
}
fn name_len(self: const Person) - integer {
    length(self.name)  // const: read-only access guaranteed
}
p.greet()     // dot syntax
greet(p)      // free-function call — identical
```

```
impl Person {
    fn greet(&self) -> String {
        format!("Hello, {}!", self.name)
    }
    fn name_len(&self) -> usize {
        self.name.len()
    }
}
p.greet();     // only dot syntax
```

Upside No impl block ceremony. Methods and free functions are the same thing — just a naming convention. Functions can be added to any type from any file without modifying the original struct definition, similar to extension methods. Mark the first parameter const to guarantee the method never modifies the receiver — analogous to Rust's &self.

Downside No consuming self. Plain (non-const) methods behave like &mut self — the caller's value is always mutably aliased. Multiple functions with the same name but different non-variant self types are a compile error — no standard overloading.

3.0.10. Polymorphic enum dispatch

```
enum Shape {
    Circle { r: float },
    Rect   { w: float, h: float }
}
fn area(self: Circle) - float { PI * pow(self.r, 2.0) }
fn area(self: Rect)   - float { self.w * self.h }

s.area()    // dispatches on the runtime variant
```

```
enum Shape {
    Circle { r: f64 },
    Rect   { w: f64, h: f64 },
}
fn area(s: &Shape) -> f64 {
    match s {
        Shape::Circle { r } => PI * r * r,
        Shape::Rect { w, h } => w * h,
    }
}
```

Upside Each variant's behaviour lives in its own small function — easy to read and extend. Adding a new type of shape only requires a new fn area(self: NewShape) without modifying the dispatch site. Feels like OOP method overriding without the inheritance.

Downside Rust's match is exhaustive: the compiler forces you to handle every variant. In loft, leaving a variant without an implementation emits a compiler warning but does not stop compilation. To suppress the warning and produce a null return, write an explicit empty-body stub: fn area(self: NewShape) -> float { }. Exhaustiveness is enforced by discipline, not the type system.

3.0.11. Closures — same-scope capture works

```
// Capture works when called in the same scope:
offset = 10;
shift = fn(x: integer) - integer { x + offset };
assert(shift(5) == 15, "shift");

// Non-capturing lambdas work with map/filter/reduce:
doubled = map([1, 2, 3], fn(x: integer) - integer { x * 2 });
evens   = filter([1, 2, 3, 4], |x| { x % 2 == 0 });

// Cross-scope: returning a capturing lambda works:
fn make_adder(n: integer) - fn(integer) - integer {
    fn(x: integer) - integer { x + n }
}
```

```
// Closure captures context freely, any scope:
let offset = 5;
let shift = |x| x + offset;
assert_eq!(shift(5), 10);

// Works as argument to higher-order functions:
let shifted: Vec<i32> = v.iter()
    .map(|x| x + offset)
    .collect();

// Returning a capturing closure:
fn make_adder(n: i32) -> impl Fn(i32) -> i32 {
    move |x| x + n
}
```

Upside Same-scope capture works for integers, text, and mutable variables — no capture modes (move vs borrow), no Fn/FnMut/FnOnce trait bounds, no lifetime annotations. Both long-form (fn(x: integer) -> integer { x * 2 }) and short-form (|x| { x * 2 }) lambdas are supported. Named function references (fn name) are compile-checked.

Downside Capture is always by value (like Rust move). Rust's Fn/FnMut/FnOnce traits give more flexibility: closures can borrow by reference, be stored in structs, and passed freely across scopes. Loft captures at definition time only — no borrow-based captures.

3.0.12. Generic functions — pass-through only, no trait bounds

```
// Pass-through generics work:
fn identity<T>(x: T) - T { x }
identity(42)        // integer
identity("hello")  // text

// Trait-bounded generics are not supported:
// fn max<T: PartialOrd>(a: T, b: T) -> T { ... }  // not valid
// Must write a version per type:
```

```
fn max_int(a: integer, b: integer) - integer {
    if a  b { a } else { b }
}
```

```
fn identity<T>(x: T) -> T { x }

fn max<T: PartialOrd>(a: T, b: T) -> T {
    if a > b { a } else { b }
}
// One function works for all comparable types.
```

Upside Basic generic functions (identity, pick_second, wrappers) work out of the box. Collections (vector<T>, hash<T>, sorted<T>) are generic at the engine level, covering the most common need. No dyn Trait, impl Trait, or associated types to learn.

Downside No trait bounds — a generic function cannot constrain T to be comparable, printable, or hashable. Any algorithm that requires an operation on T (comparison, addition, display) must be written once per concrete type. There is no way for user code to define "anything sortable" without duplicating the sort logic.

3.0.13. String formatting — embedded expressions

```
msg  = "Hi {name}, score: {score:+6.2}"  // width.prec
prec = "{value:.2}"                      // precision only
hex  = "{n:#x}"
list = "{for x in 1..4 {x*2}}"          // "2,4,6"
     = "{\"hello\":3}"                    // nested string literal — works
```

```
let msg  = format!("Hi {name}, score: {:+6.2}", score);
let prec = format!("{:.2}", value);
let hex  = format!("{:#x}", n);
// no in-string iteration; needs a separate collect step
// nested string literals in format args are fine in Rust
```

Upside Concise — no format!() call, no separate variable. Full format expressions (arithmetic, slices, for comprehensions) can appear directly inside {}. Specifiers mirror Rust: :width, :.precision, :width.precision, sign (+), radix (#x, #o, b), alignment (<, >, ^), and zero-padding (0N) all work.

Downside Unknown radix letters in specifiers are compile-time errors (e.g. :5z or :5B are both rejected). Radix letters are case-sensitive: valid ones are b, o, x/X, e, and j/json — uppercase B or O produce an error. Applying a numeric specifier to an incompatible type (such as :x on a text value, or zero-padding on a boolean) is a compile-time error.

3.0.14. Built-in parallel for-loops — par(...)

```
fn double(r: const Score) - integer { r.value * 2 }

sum = 0;
```

```
for item in scores par(b=double(item), 4) {
    sum += b;   // b is double(item), computed in parallel
}               // results arrive in original order

// Method form:
for item in scores par(b=item.value(), 4) { ... }
```

```
use rayon::prelude::*;

fn double(r: &Score) -> i32 { r.value * 2 }

let sum: i32 = scores.par_iter()
    .map(double)
    .sum();      // order is not guaranteed without extra work

// rayon is an external crate; add to Cargo.toml first
```

Upside Built into the language — no external crate, no Cargo.toml edit. The compiler validates the worker function signature at the call site. Results are delivered in the original vector order. The thread count is set per call, making it easy to tune for the hardware at hand.

Downside Workers can return primitives (integer, long, float, boolean), text, and inline enums — but not struct references. Workers cannot capture local variables: context must be embedded as fields alongside the data in the element struct. For multi-stage transformations, use map() / filter() / reduce() sequentially — each stage allocates a new intermediate vector.

4. vs Python

Loft and Python share a similar surface syntax — assignment without type annotations, brace-free expressions, and names that just work. But loft is statically typed and compiled to bytecode, while Python is dynamically typed and interpreted. This page documents the most important differences so that Python programmers know exactly what they gain and what they give up.

4.0.1. Variables — static types inferred at first assignment

```
x = 42           // integer — fixed at first assignment
x += 1
name = "Bob"     // text — a different variable, not a rebind
x = "oops"      // compile error: cannot assign text to integer
```

```
x = <span class="nm">42</span>           <span class="cm">># int inferred; can be
rebound to any type</span>
x += <span class="nm">1</span>
name = <span class="st">"Bob"</span>
x = <span class="st">"oops"</span>       <span class="cm">># fine — x is now a
str</span>
```

Upside Type errors are caught before the program runs. Renaming or reshaping data structures surfaces mismatches immediately. There is no equivalent of Python's runtime TypeError, AttributeError, or "NoneType has no attribute X" crash — those classes of bug are detected at compile time.

Downside Types are fixed at first assignment and cannot change. Python's dynamic typing genuinely speeds up prototyping: a function that accepts anything, a list that holds mixed types, or a variable that starts as an integer and becomes a float later are all natural in Python and impossible in loft. Adding a type annotation layer (mypy, pyright) gives Python static safety without giving up its flexibility.

4.0.2. Null — structured absence, not a universal wildcard

```
struct User {
    email: text,             // nullable by default
    age:   integer not null,  // can never be null
}
u = User { age: 30 };
if u.email == null { print("no email"); }
u.age = null;  // compile error: age is not null
```

```
<span class="kw">from</span> dataclasses <span class="kw">import</span> dataclass
<span class="kw">from</span> typing <span class="kw">import</span> Optional

<span class="en">@dataclass</span>
<span class="kw">class</span> <span class="en">User</span>:
    age: <span class="bi">int</span>
    email: Optional[<span class="bi">str</span>] = <span class="kw">None</span>
<span class="cm">># explicit Optional</span>
```

```
u = <span class="fn-call">User</span>(age=<span class="nm">30</span>)
<span class="kw">if</span> u.email <span class="kw">is</span> <span
class="kw">None</span>:
    <span class="bi">print</span>(<span class="st">"no email"</span>)
u.age = <span class="kw">None</span>  <span class="cm"># allowed at runtime; mypy
catches this</span>
```

Upside Nullability is part of the struct definition — readable at a glance. not null fields are enforced by the compiler and additionally locked at runtime in debug builds, catching accidental null writes early in development. No Optional[T] wrapping needed; the comparison v == null is natural.

Downside The default is nullable, not non-nullable — the safe default is backwards from what static analysis advocates recommend. Python with mypy and Optional[T] annotation gives static guarantees that loft's runtime checks do not. Python's None also participates in truthiness testing (if not email:), pattern matching, and or chaining in ways that loft's null does not support.

4.0.3. Structs vs classes and dicts

```
struct Point { x: float, y: float }
p = Point { x: 1.0, y: 2.0 };
p.x += 0.5;
p.z;          // compile error: no field z on Point
p.x = "hi"; // compile error: cannot assign text to float

fn distance(self: const Point) - float {
    sqrt(self.x * self.x + self.y * self.y)
}
```

```
<span class="kw">from</span> dataclasses <span class="kw">import</span> dataclass
<span class="kw">import</span> math

<span class="en">@dataclass</span>
<span class="kw">class</span> <span class="en">Point</span>:
    x: <span class="bi">float</span>
    y: <span class="bi">float</span>

    <span class="kw">def</span> <span class="fn-call">distance</span>(self) ->
<span class="bi">float</span>:
        <span class="kw">return</span> math.<span class="fn-call">sqrt</
span>(self.x**<span class="nm">2</span> + self.y**<span class="nm">2</span>)

p = <span class="fn-call">Point</span>(x=<span class="nm">1.0</span>, y=<span
class="nm">2.0</span>)
p.z        <span class="cm">># AttributeError at runtime (or dict: KeyError)</
span>
p.x = <span class="st">"hi"</span> <span class="cm">># allowed at runtime; mypy
catches this</span>
```

Upside Field access is checked at compile time — typos in field names are caught before any code runs. Struct memory layout is fixed and unboxed; integer and float fields live directly in memory with no heap allocation overhead. Methods are ordinary named functions — they can be added from any file, at any time, without modifying the struct definition.

Downside No inheritance, no __repr__, no operator overloading (__add__, __eq__, etc.), no properties or descriptors. Python's @dataclass generates __init__, __repr__, and __eq__ automatically. Loft structs are data holders; all display and comparison logic must be written by hand. Python also supports plain dicts as lightweight records, which is often more convenient for ad-hoc data.

4.0.4. while loop — yes; no loop or while … else

```
while !ready() { step(); }  // works

while length(queue)  0 {
    process(queue[0]);
    queue#remove;
}

// No loop keyword — infinite loop needs a large bound:
for _ in 0..2147483647 {
    if should_exit() { break; }
    step();
}
```

```
<span class="kw">while</span> <span class="kw">not</span> <span class="fn-
call">ready</span>():
    <span class="fn-call">step</span>()

<span class="kw">while</span> queue:
    <span class="fn-call">process</span>(queue.<span class="fn-call">pop</
span>(<span class="nm">0</span>))

<span class="kw">while</span> <span class="kw">True</span>:          <span
class="cm"># truly infinite</span>
    <span class="kw">if</span> <span class="fn-call">should_exit</span>(): <span
class="kw">break</span>
    <span class="fn-call">step</span>()
```

Upside while condition { } works exactly as expected. Filtered loops (for x in v if pred(x)) and loop attributes (x#first, x#count) add expressive power to for without needing a separate loop form.

Downside No while True: equivalent — infinite loops require a bounded for range as a workaround. No while … else (executes when the condition becomes false without a break), a Python pattern with no clean equivalent in loft.

4.0.5. Polymorphic enum dispatch vs isinstance

```
enum Shape {
    Circle { r: float },
```

```
    Rect   { w: float, h: float }
}
fn area(self: Circle) - float { PI * pow(self.r, 2.0) }
fn area(self: Rect)   - float { self.w * self.h }

s.area()   // dispatches on the runtime variant
```

```
<span class="kw">import</span> math
<span class="kw">from</span> dataclasses <span class="kw">import</span> dataclass

<span class="en">@dataclass</span>
<span class="kw">class</span> <span class="en">Circle</span>: r: <span
class="bi">float</span>
<span class="en">@dataclass</span>
<span class="kw">class</span> <span class="en">Rect</span>:   w: <span
class="bi">float</span>; h: <span class="bi">float</span>

<span class="kw">def</span> <span class="fn-call">area</span>(s):
    <span class="kw">if</span> <span class="bi">isinstance</span>(s, Circle):
<span class="kw">return</span> math.pi * s.r ** <span class="nm">2</span>
    <span class="kw">if</span> <span class="bi">isinstance</span>(s, Rect):
<span class="kw">return</span> s.w * s.h

<span class="cm"># Python 3.10+ structural pattern matching:</span>
<span class="kw">match</span> s:
    <span class="kw">case</span> <span class="en">Circle</span>(r=r): <span
class="kw">return</span> math.pi * r ** <span class="nm">2</span>
    <span class="kw">case</span> <span class="en">Rect</span>(w=w, h=h): <span
class="kw">return</span> w * h
```

Upside Each variant's behaviour lives in its own small, named function — easy to read, easy to navigate in an editor. Adding a new shape only requires a new fn area(self: NewShape) with no changes to existing code or a central dispatch function. The compiler warns when a variant has no implementation for a called method.

Downside Unlike Rust's match, loft does not enforce exhaustiveness — a missing variant implementation produces only a warning, not a compile error. Python 3.10+ structural pattern matching (match/case) fully destructs the matched value and is exhaustive when case _: is present. Python also supports inheritance-based polymorphism (class Circle(Shape)) and abstract base classes, giving far richer dispatch options.

4.0.6. String formatting — embedded expressions

```
msg  = "Hi {name}, score: {score:+8.2}"
hex  = "{n:#x}"
list = "{for x in 1..4 {x*2}}"   // "2,4,6"
pad  = "{value:08}"               // zero-padded
```

```
msg  = <span class="st">f"Hi {name}, score: {score:+8.2f}"</span>
hex  = <span class="st">f"{n:#x}"</span>
<span class="cm">># no loop in f-string; use a join:</span>
lst  = <span class="st">","</span>.<span class="fn-call">join</span>(<span
class="bi">str</span>(x*<span class="nm">2</span>) <span class="kw">for</span> x
<span class="kw">in</span> <span class="bi">range</span>(<span class="nm">1</
span>, <span class="nm">4</span>))  <span class="cm">># "2,4,6"</span>
pad  = <span class="st">f"{value:08}"</span>
```

Upside All loft strings are implicitly format strings — no f prefix needed. Inline for loops inside {} produce comma-separated output without a separate join. Format specifiers mirror Python's f-string mini-language: width, precision, sign, alignment, zero-padding, and radix (#x, #o, b) all work.

Downside Python f-strings accept arbitrary expressions: method calls ({obj.method():.2f}), ternary expressions ({"yes" if ok else "no"}), and join operations inline. Loft restricts what can appear inside {}. Python also supports conversion flags (!r for repr, !s for str, !a for ASCII) and the = debug specifier ({x=} prints x=42); loft has none of these.

4.0.7. Collections — typed and built-in

```
nums:   vector<integer> = [1, 2, 3];
lookup: hash<text>    = {};
lookup["key"] = "value";
scores: sorted<integer> = {};
scores[user] = 95;  // O(log n) keyed insert

// Element type is enforced at compile time:
nums += ["x"];  // compile error: expected integer
```

```
nums   = [<span class="nm">1</span>, <span class="nm">2</span>, <span
class="nm">3</span>]          <span class="cm">># list — any element type</span>
lookup = {}                 <span class="cm">># dict — any key/value type</span>
lookup[<span class="st">"key"</span>] = <span class="st">"value"</span>

<span class="cm">># sorted dict requires an external package:</span>
<span class="kw">from</span> sortedcontainers <span class="kw">import</span>
SortedDict
scores = <span class="fn-call">SortedDict</span>()
scores[user] = <span class="nm">95</span>

nums.<span class="fn-call">append</span>(<span class="st">"x"</span>)
<span class="cm">># allowed at runtime</span>
```

Upside All collection types — vector, hash, and sorted map — are built in; no extra import or install needed. Element types are checked at compile time. The same [] indexing syntax works on all three. for x in v if pred(x) { v#remove; } safely removes the current element while iterating — something Python requires careful index management for.

Downside Python's built-in dict and list are among the most heavily optimised data structures in any scripting runtime. Python also has set and frozenset (no loft equivalent), ordered insertion semantics on dict (Python 3.7+), and an enormously expressive comprehension syntax ([x*2 for x in v if x > 0]). Loft's map() / filter() / reduce() higher-order functions allocate a new vector at each stage, while Python's generators are lazy and allocation-free until consumed.

4.0.8. Closures — same-scope capture works

```
// Capture works when called in the same scope:
offset = 10;
shift = fn(x: integer) - integer { x + offset };
assert(shift(5) == 15, "shift");

// Non-capturing lambdas work with map/filter/reduce:
doubled = map([1, 2, 3], fn(x: integer) - integer { x * 2 });
evens   = filter([1, 2, 3, 4], |x| { x % 2 == 0 });

// Capturing lambda with map:
offset = 10;
shifted = map(nums, fn(x: integer) - integer { x + offset });
```

```
offset = <span class="nm">10</span>
shift = <span class="kw">lambda</span> x: x + offset  <span class="cm"># capture
works anywhere</span>
<span class="kw">assert</span> <span class="fn-call">shift</span>(<span
class="nm">5</span>) == <span class="nm">15</span>

nums = [<span class="nm">1</span>, <span class="nm">2</span>, <span
class="nm">3</span>, <span class="nm">4</span>, <span class="nm">5</span>]
doubled = <span class="bi">list</span>(<span class="bi">map</span>(<span
class="kw">lambda</span> x: x * <span class="nm">2</span>, nums))
evens   = <span class="bi">list</span>(<span class="bi">filter</span>(<span
class="kw">lambda</span> x: x % <span class="nm">2</span> == <span class="nm">0</span>
span>, nums))

<span class="cm">># capturing lambda passed to map:</span>
shifted = [x + offset <span class="kw">for</span> x <span class="kw">in</span>
nums]
```

Upside Same-scope capture works for integers, text, and mutable variables — no capture modes, no scope surprises. Both long-form (fn(x: integer) -> integer { x * 2 }) and short-form (|x| { x * 2 }) lambdas are supported. Named function references (fn name) are compile-checked. Higher-order functions (map, filter, reduce) accept both lambdas and fn-refs.

Downside Capture is by value at definition time — later mutations to the original variable do not affect the lambda (and vice versa). Python's lambda and nested def close over variables by reference, so mutations are shared. Python's list comprehensions ([x + offset for x in v]) are shorter than map(v, fn(x) { x + offset }).

4.0.9. No exception handling — file errors use FileResult

```
// Logic errors: assert aborts the program
fn divide(a: float, b: float) - float {
    assert(b != 0.0, "division by zero");
    a / b
}

// Reading: check f#exists before using content
f = file("data.txt");
if f#exists {
    data = f.content();
} else {
    print("file not found");
}

// Mutating ops return a FileResult enum
result = delete("old.txt");
match result {
    FileResult.Ok             - print("deleted");
    FileResult.NotFound       - print("not found");
    FileResult.PermissionDenied - print("permission denied");
    FileResult.IsDirectory    - print("is a directory");
    _                         - print("other error");
}

// Or just check success:
if !move("a.txt", "b.txt").ok() { print("rename failed"); }
```

```
<span class="cm"># Logic errors: raise an exception</span>
<span class="kw">def</span> <span class="fn-call">divide</span>(a: <span
class="bi">float</span>, b: <span class="bi">float</span>) -> <span
class="bi">float</span>:
    <span class="kw">if</span> b == <span class="nm">0</span>:
        <span class="kw">raise</span> <span class="fn-call">ValueError</
span>(<span class="st">"division by zero"</span>)
    <span class="kw">return</span> a / b

<span class="kw">try</span>:
    result = <span class="fn-call">divide</span>(x, y)
<span class="kw">except</span> ValueError <span class="kw">as</span> e:
    <span class="bi">print</span>(<span class="st">f"error: {e}"</span>)

<span class="cm"># File errors: catch specific exception types</span>
<span class="kw">try</span>:
    <span class="kw">with</span> <span class="bi">open</span>(<span
class="st">"data.txt"</span>) <span class="kw">as</span> f:
        data = f.<span class="fn-call">read</span>()
<span class="kw">except</span> FileNotFoundError:
    <span class="bi">print</span>(<span class="st">"file not found"</span>)
```

```
<span class="kw">try</span>:
    os.<span class="fn-call">remove</span>(<span class="st">"old.txt"</span>)
<span class="kw">except</span> FileNotFoundError:
    <span class="bi">print</span>(<span class="st">"not found"</span>)
<span class="kw">except</span> PermissionError:
    <span class="bi">print</span>(<span class="st">"permission denied"</span>)
<span class="kw">except</span> IsADirectoryError:
    <span class="bi">print</span>(<span class="st">"is a directory"</span>)
```

Upside File errors are represented as a typed FileResult enum — Ok, NotFound, PermissionDenied, IsDirectory, NotDirectory, Other — so every failure case is named and exhaustively matchable. No accidental exception swallowing, no hidden exit paths, no stack-unwinding overhead. The control flow of every loft function is straightforward to read.

Downside Logic errors (assert, panic) abort the entire program — there is no try/except to catch them. Python's exception hierarchy supports retry logic, cleanup via finally, and graceful degradation from arbitrary errors anywhere in the call stack — patterns that are not expressible in loft today.

4.0.10. Built-in parallel for-loops — par(...)

```
fn score(item: const Record) - integer { item.value * 2 }

total = 0;
for item in records par(s=score(item), 4) {
    total += s;   // results arrive in original order
}                 // 4 worker threads, no GIL
```

```
<span class="kw">from</span> concurrent.futures <span class="kw">import</span>
ProcessPoolExecutor

<span class="kw">def</span> <span class="fn-call">score</span>(item): <span
class="kw">return</span> item.value * <span class="nm">2</span>

<span class="cm"># ProcessPoolExecutor: bypasses GIL via separate processes</
span>
<span class="kw">with</span> <span class="fn-call">ProcessPoolExecutor</
span>(max_workers=<span class="nm">4</span>) <span class="kw">as</span> ex:
    results = <span class="bi">list</span>(ex.<span class="fn-call">map</
span>(score, records))
total = <span class="bi">sum</span>(results)

<span class="cm">># ThreadPoolExecutor: simpler but GIL limits CPU-bound work</
span>
```

Upside Built into the language — no import, no boilerplate. The GIL does not apply; worker threads run on separate OS threads inside the same process. Results arrive in the original vector order. The compiler validates the worker function signature at the call site. The thread count is set per call, making it easy to tune for the hardware.

Downside Workers can return primitives (integer, long, float, boolean), text, and inline enums — but not struct references. Context must be embedded as fields in the element struct; workers cannot capture local variables. Python's ProcessPoolExecutor works with any picklable object and gives full control over timeouts, cancellation, and error propagation.

4.0.11. Generic functions — pass-through only, no duck typing

```
// Pass-through generics work:
fn identity<T>(x: T) - T { x }
identity(42)        // integer
identity("hello")  // text

// Algorithms requiring operations on T still need per-type versions:
fn max_int(a: integer, b: integer) - integer {
    if a  b { a } else { b }
}
```

```
<span class="cm"># Duck typing: works for any T that supports ></span>
<span class="kw">def</span> <span class="fn-call">max_val</span>(a, b):
    <span class="kw">return</span> a <span class="kw">if</span> a > b <span
class="kw">else</span> b

<span class="cm">># With type hints (Python 3.12+):</span>
<span class="kw">from</span> typing <span class="kw">import</span> TypeVar
T = <span class="fn-call">TypeVar</span>(<span class="st">"T"</span>)
<span class="kw">def</span> <span class="fn-call">max_val</span>(a: T, b: T) ->
T:
    <span class="kw">return</span> a <span class="kw">if</span> a > b <span
class="kw">else</span> b
```

Upside Basic generic functions (pass-through, wrapping, forwarding) work out of the box. Collections (vector<T>, hash<T>, sorted<T>) are generic at the engine level, covering the most common need without any user-visible type parameter syntax.

Downside A generic function cannot perform operations on T — no comparison, no arithmetic, no display. Any algorithm that requires an operation must be written once per concrete type. Python's duck typing lets a single function work on any type that supports the needed operation (len, >, +) without explicit annotation. Python 3.12 TypeVar and Protocol add optional static checking on top.

4.0.12. Function signatures — defaults and named args, but no *args

```
fn connect(host: text, port: integer = 80, tls: boolean = true) - text { ... }

connect("localhost");                // uses defaults
connect("localhost", tls: false);    // named arg — skips port
connect("localhost", 443, false);   // positional — all explicit
```

```
<span class="kw">def</span> <span class="fn-call">connect</span>(host: <span
class="bi">str</span>, port: <span class="bi">int</span> = <span class="nm">80</
```

```
span>, tls: <span class="bi">bool</span> = <span class="kw">True</span>):
    ...

<span class="fn-call">connect</span>(<span class="st">"localhost"</span>)
<span class="cm"># uses defaults</span>
<span class="fn-call">connect</span>(<span class="st">"localhost"</span>,
tls=<span class="kw">False</span>) <span class="cm">># keyword arg — skips port</
span>

<span class="kw">def</span> <span class="fn-call">log</span>(*args,
**kwargs): ...   <span class="cm">># variadic</span>
```

Upside Default parameter values and named arguments work similarly to Python. connect("host", tls: false) skips defaulted middle parameters. Every call site is readable without jumping to the function definition.

Downside No *args or **kwargs — variadic dispatch must use a vector parameter. Python's flexible argument syntax also enables decorator patterns and configuration-driven code that is harder to express in loft.

4.0.13. Ecosystem — minimal standard library

```
// import "mylib" loads a .loft file relative to the script.
// The full standard library ships inside the interpreter binary;
// there is no package manager or external dependency system.

// Built-in: text, math, file I/O, collections,
//           logging, threading, image, lexer/parser
```

```
<span class="kw">import</span> numpy <span class="kw">as</span> np          <span
class="cm">># numerical arrays / SIMD</span>
<span class="kw">import</span> pandas <span class="kw">as</span> pd         <span
class="cm">># dataframes</span>
<span class="kw">import</span> requests             <span class="cm">># HTTP</
span>
<span class="kw">import</span> flask                <span class="cm">># web
framework</span>
<span class="kw">import</span> sqlalchemy           <span class="cm">># database
ORM</span>
<span class="kw">import</span> scikit_learn         <span class="cm">># machine
learning</span>
<span class="cm">># 500 000+ packages on PyPI, installable with:</span>
<span class="cm">#   pip install <package></span>
```

Upside Zero external dependencies — the interpreter is a single self-contained binary. Deployment is copying one file. There is no requirements.txt, no virtual environment, no version conflict, and no supply-chain risk. The built-in library covers text manipulation, math, file I/O, typed collections, parallel execution, image handling, and a full lexer/parser framework.

Downside Python's ecosystem is its defining advantage. NumPy, pandas, scikit-learn, TensorFlow, requests, Flask, SQLAlchemy, pytest, and hundreds of thousands of other packages are not available to loft programs. Any data science, web development, database integration, or protocol implementation task will require reimplementing from scratch what Python solves with a single pip install. For these domains, loft is the wrong tool today.

4.0.14. Exponentiation uses pow(); ^ is XOR

```
area  = PI * pow(r, 2.0)  // exponentiation via pow()
bits  = a | b              // bitwise OR
xor   = a ^ b              // bitwise XOR — not exponentiation
cube  = pow(x, 3.0)       // cube root: pow(x, 1.0/3.0)
```

```
<span class="kw">import</span> math
area  = math.pi * r ** <span class="nm">2</span>   <span class="cm"># ** is
exponentiation</span>
bits  = a | b              <span class="cm"># bitwise OR</span>
xor   = a ^ b              <span class="cm"># bitwise XOR</span>
cube  = x ** <span class="nm">3</span>            <span class="cm"># integer
cube — exact</span>
cube  = x ** (<span class="nm">1</span>/<span class="nm">3</span>)         <span
class="cm"># cube root as float</span>
```

Upside ^ behaves as bitwise XOR in both loft and Python — no mismatch. Bitwise operator precedence matches: | (loose) → ^ → & → shifts → arithmetic (tight). The pow() function is explicit, avoiding any ambiguity about whether ^ means XOR or exponentiation.

Downside Python's ** operator works on integers, floats, and complex numbers and produces the exact type the operands imply. Loft's pow() operates on float and single only — integer exponentiation has no built-in; compute it with a loop or cast to float first. Python's pow(base, exp, mod) three-argument form computes modular exponentiation efficiently; loft offers no equivalent.

5. Keywords

This page covers the building blocks that shape how your program makes decisions and repeats work: conditions, loops, breaks, and a few loop helpers that make common patterns shorter. Every example is verified with an assert so you can see the exact expected result.

```
fn main() {
```

5.0.1. Conditionals

'if' runs a block only when its condition is true. If the condition is false the block is skipped entirely — nothing else happens. 'panic' stops the program immediately with a message. During development it is a clear way to mark situations that should never occur.

```
  if 2 > 5 {
    panic("Incorrect test");
  }
```

Combine conditions with 'and' / 'or' (or their symbol equivalents '&&' / '||'). Note that '&' is the bitwise AND operator — it works on the individual 1s and 0s inside a number, which is different from the logical 'and' keyword that works on true/false values. An 'if/else' chain picks exactly one branch to execute.

```
  a = 12;
  if a > 10 and a & 7 == 4 {
    a += 1;
  } else {
    a -= 1;
  }
```

Text enclosed in double quotes can contain expressions inside curly braces — the expression is evaluated and its result is inserted into the text. 'assert' checks that its first argument is true; if it is false the second argument is printed as an error message.

```
  assert(a == 13, "Incorrect value {a} != 13");
```

5.0.2. if as an expression

Every block in Loft produces a value — the last expression inside it. That means you can use 'if/else' on the right-hand side of an assignment, picking between two values based on a condition. No ternary operator needed.

```
  b = if a == 13 {
    "Correct"
  } else {
    "Wrong"
  };
  assert(b == "Correct", "Logic expression");
```

5.0.3. Iteration

'for x in a..b' loops over the integers starting at a and stopping before b (exclusive upper bound). Use '..=' to include the upper bound. Here we sum 1 + 2 + 3 + 4 + 5 = 15.

```
t = 0;
for a in 1..6 {
  t += a;
}
assert(t == 15, "Total was {t} instead of 15");
```

'rev(range)' reverses the direction of any range. '1..=5' is inclusive, so it visits 5, 4, 3, 2, 1 in that order. Multiplying each digit into a running total builds the number 54321.

```
t = 0;
for a in rev(1..=5) {
  t = t * 10 + a;
}
assert(t == 54321, "Result was {t} instead of 54321");
```

5.0.4. Nested loops and break

Loft has no 'while' or 'loop' keyword — all repetition uses 'for' with a range, so there is always a clear upper bound on how many iterations can occur. 'break' exits the nearest enclosing loop immediately. To exit an outer loop from inside an inner one, write 'outerVar#break'. Here x#break leaves both loops when the product x*y reaches 16.

```
b = "";
for x in 1..5 {
  for y in 1..5 {
    if y > x {
      break;
```

this breaks the inner y loop

```
    }
    if x * y >= 16 {
      x# break;
    }
    if len(b) > 0 {
      b += "; ";
    }
    b += "{x}:{y}";
  }
}
assert(b == "1:1; 2:1; 2:2; 3:1; 3:2; 3:3; 4:1; 4:2; 4:3", "Incorrect sequence
'{b}'");
```

5.0.5. Loop helpers: #first and #count

Inside a loop body you have access to some useful metadata about the current iteration. ‘x#first’ is true only for the very first element that passes the filter. ‘x#count’ is a zero-based index counting only the elements that were not filtered out. An ‘if’ clause directly after ‘in range’ filters which values enter the loop — here we skip every value where x % 3 == 1, keeping 2, 3, 5, 6, 8, 9.

```
  b = "";
  for x in 1..=9 if x % 3 != 1 {
    if !x#first {
      b += ", ";
    }
    b += "{x#count}:{x}";
  }
  assert(b == "0:2, 1:3, 2:5, 3:6, 4:8, 5:9", "Sequence '{b}'");
```

5.0.6. Formatting a loop inline

A for loop can appear inside a format string. The results are collected into a bracketed, comma-separated list. A format specifier after the closing brace is applied to every element — ‘:02’ means at least 2 digits, zero-padded. This is handy for building compact representations on the fly.

```
  assert("a{for x in 1..7 {x*2}:02}b" == "a[02,04,06,08,10,12]b", "Format
range");
}
```

6. Texts

Text is one of the most frequently used types in any real program — for output, for reading input, for building messages. This page shows you how Loft stores and manipulates text, how to measure it, slice it, search it, and format it exactly the way you need.

The key thing to know upfront: Loft stores text as a sequence of bytes using UTF-8 encoding — the same format used on the web. Plain ASCII letters, digits, and punctuation each take exactly one byte. Accented letters, emoji, and many non-Latin scripts take 2–4 bytes. Most operations count bytes rather than visible characters. That makes them fast and simple, but you need to keep multi-byte characters in mind when you slice by position.

```
fn main() {
```

6.0.1. Joining and measuring text

Use '+' to join two pieces of text into one. The result is a new value — neither of the originals is modified. Placing a value name inside curly braces inside a format string inserts the value as text. You can also control the width: '{a:4}' pads the value on the right to at least 4 characters.

```
  a = "1" + "2";
  assert("'{a:4}'" == "'12  '", "Formatting text");
  assert(len(a + "123") == 5, "Text length");
```

'len()' counts bytes, not visible characters. An emoji like '😄' takes 4 bytes in UTF-8, a heart symbol '♥' takes 3, and every plain ASCII letter takes exactly 1. Keep this in mind whenever you use len() to check how "long" something is to a human reader.

```
  assert(len("😄") == 4, "Emoji byte length");
  assert(len("♥") == 3, "Heart byte length");
  assert(len("abc") == 3, "ASCII byte length");
```

6.0.2. Reading individual characters

Square brackets with a single byte index give you the character at that position. For plain ASCII text every byte is one character, so indexing by position is straightforward. Loft will return the full character even if it spans multiple bytes.

```
  s = "ABCDE";
  assert(s[0] == 'A', "Character at byte 0");
  assert(s[2] == 'C', "Character at byte 2");
```

6.0.3. Slicing text

A slice 's[start..end]' gives you the bytes from 'start' up to but not including 'end'. You can omit the start to begin at 0, or omit the end to go all the way to the last byte. Loft snaps offsets to valid character boundaries so you can never accidentally cut a multi-byte character in half.

```
assert(s[0..2] == "AB", "Explicit start slice");
assert(s[..2] == "AB", "Open-start slice");
assert(s[1..4] == "BCD", "Sub-string by byte range");
assert(s[3..] == "DE", "Open-ended sub-string");
```

A negative end index counts backwards from the end of the text. '−1' means "stop one byte before the very last byte". Combined with a start offset this lets you trim a known suffix without knowing the exact length.

```
txt = "12😊☹45";
assert(txt[2..- 1] == "😊☹4", "UTF-8 sub-string by byte range");
```

6.0.4. Iterating over characters

A 'for' loop over a text value visits one character at a time, even when characters span multiple bytes. Two loop helpers give you position information without any extra code: 'c#index' is the byte offset where the current character starts. 'c#next' is the byte offset immediately after the current character ends. This makes it easy to build a byte-offset map or to collect characters starting from a specific position.

```
result = "";
positions =[];
nexts =[];
for c in "Hi 😊!" {
  positions +=[c#index];
  nexts +=[c#next];
  if c#index >= 3 {
    result += c;
  }
}
assert(result == "😊!", "Character iteration was {result}");
assert("{positions}" == "[0,1,2,3,7]", "Character positions was {positions}");
assert("{nexts}" == "[1,2,3,7,8]", "Character nexts was {nexts}");
```

6.0.5. Searching inside text

These built-in functions answer common "does this text contain…?" questions. 'starts_with' and 'ends_with' check the boundaries. 'find' returns the byte offset of the first match, or null if not found. 'contains' is true if the needle appears anywhere in the text. All positions are byte offsets, consistent with 'len()' and slicing.

```
assert("something".starts_with("some"), "starts_with");
assert("something".ends_with("thing"), "ends_with");
assert("something".find("th") == 4, "find returns byte offset");
assert("a longer text".contains("longer"), "contains");
```

'trim()' removes spaces and other whitespace from both ends of a text value. It is useful when reading user input or parsing text from a file.

```
  assert(trim("  hello  ") == "hello", "trim");
```

6.0.6. Escaping braces in format strings

Inside a format string '{' and '}' are special: they introduce an interpolated expression. To include a literal brace in the output, double it: '{{' produces '{' and '}}' produces '}'.

```
  brace_inner = "cd";
  assert("ab{{cd}}e" == "ab{{{brace_inner}}}e", "Escaping braces");
```

6.0.7. Aligning text in a fixed-width field

The ':' specifier controls alignment and width. '<' left-aligns the value, '>' right-aligns it (the default). The width can be a constant or a small arithmetic expression — here '2+3' evaluates to 5, giving a field of width 5.

```
  vr = "abc";
  assert("1{vr:<2+3}2{vr}3{vr:6}4{vr:>7}" == "1abc  2abc3abc   4    abc", "Text
alignment");
}
```

7. Integers

Numbers are at the heart of almost every program. This page covers the integer types Loft provides, the arithmetic and bitwise operations you can perform on them, how to convert between numbers and text, and what Loft does when something goes wrong — such as dividing by zero.

The default number type is 'integer': a 32-bit signed whole number, the same as i32 in Rust. It can hold values from about −2 billion to +2 billion. For larger values Loft also has 'long' (64-bit), shown at the end of this page. For decimal (fractional) numbers, see the Float page.

```
fn main() {
```

7.0.1. Converting between numbers and text

Wrapping a value in '{...}' inside a string formats it as text. Going the other way, 'as integer' parses a text value into a number. If the text cannot be parsed, the result is null — not a crash.

```
  v = 4;
  assert("{v}" == "4", "Format integer as text");
  assert("123" as integer == 123, "Parse text to integer");
  assert(!("abc" as integer), "Unparseable text gives null");
```

7.0.2. Arithmetic and operator precedence

Loft follows standard mathematical precedence: '*' and '/' before '+' and '-'. Bitwise operators (<<, &, ^) have their own precedence — when mixing them with arithmetic, parentheses make your intent clear and avoid surprises. Note: '^' is XOR, not exponentiation. Use pow(base, exp) for powers.

```
  assert(1 + 2 * 4 == 9, "Multiplication before addition");
  assert(1 + 2 << 2 == 12, "Shift: (1+2) << 2 = 12");
  assert(0x0a8 & 15 == 8, "Bitwise AND masks low 4 bits");
  assert(42 ^ 0b111111 == 21, "Bitwise XOR");
  assert(105 % 100 == 5, "Modulus (remainder)");
  assert(pow(2.0, 3.0) == 8.0, "pow() for exponentiation");
```

'abs()' returns the absolute value — the distance from zero, always positive.

```
  assert(1 + abs(-2) == 3, "abs(-2) == 2");
```

7.0.3. Division by zero produces null, not a crash

Most languages crash or throw an exception on division by zero. Loft produces null instead, which behaves like false in conditions. This lets you handle missing or bad data gracefully with a simple '!'.

```
  a = 2 * 2;
  a -= 4;
```

a is now 0

```
  assert(!(12 / a), "Division by zero gives null");
```

7.0.4. Warning when you write a literal zero divisor

When the divisor is a literal 0 written directly in your source code, loft warns you while reading your code (before running it), because that is almost certainly a mistake:

```
n / 0   // warning: Division by constant zero
n % 0   // warning: Modulo by constant zero
```

The expression still runs and returns null — the warning is informational, not an error. Use a variable (like 'a' above) when you intentionally want null-on-zero division without a warning.

7.0.5. Embedding integers in text

```
assert("a{12}b" == "a12b", "Integer in format string");
```

A full expression can appear inside '{...}', not just a variable name.

```
assert("a{1 + 2 * 3}b" == "a7b", "Expression in format string");
```

7.0.6. Number format specifiers

After a ':' inside '{...}' you can control how a number is displayed:

```
#x  — hexadecimal with '0x' prefix
o   — octal
b   — binary
+   — always show a sign (+ or -)
N   — minimum field width (space-padded on the left)
0N  — minimum field width (zero-padded on the left)
```

```
assert("a{1+2+32:#x}b" == "a0x23b", "Hex format with 0x prefix");
assert("{12:o}" == "14", "Octal");
assert("{12:+4}" == " +12", "Sign and width");
assert("{1:03}" == "001", "Zero-padded width");
assert("{42:b}" == "101010", "Binary");
```

Hexadecimal literals in source code accept both lower and upper case digits.

```
assert(0xff == 255, "Lowercase hex literal");
assert(0xFF == 255, "Uppercase hex literal");
assert(0x2A == 42, "Uppercase hex digit");
```

7.0.7. The 'long' type for large numbers

'long' is a 64-bit signed integer — use it when values might exceed 2 billion. Write a long literal by appending 'l': '1l', '100000000000l'. Long values support the same arithmetic and format specifiers as integer. Convert a long back to an integer with 'as integer'.

```
big = 10000000000l * 5l;
assert(big == 50000000000l, "Long arithmetic");
```

```
assert("{1l + 1:+4}" == "  +2", "Long in format string");
assert(12l as integer == 12, "Long to integer conversion");
```

7.0.8. Common pitfall: integer overflow

If a 32-bit integer calculation overflows the max value (2 billion), the result wraps around silently. Use ‘long’ when you expect large values. For example, a score that multiplies two large numbers can silently wrap. Switching to ‘long’ avoids this: ‘score = big_a as long * big_b as long’.

```
}
```

8. Boolean

A boolean value is either 'true' or 'false'. Booleans appear naturally wherever you make a decision: in 'if' conditions, loop filters, and comparisons. Loft adds a third state called 'null' — meaning "no value" — which behaves like false whenever a boolean is expected.

This design means you rarely need to write a separate null-check: '!x' is true both when x is false and when x is null.

```
fn main() {
```

A comparison produces a boolean result directly. '!' flips a boolean: true ⟶ false, false ⟶ true.

```
  assert(!(3 > 2 + 4), "3 is not greater than 6");
  assert(true as text == "true", "Convert boolean to text");
```

8.0.1. Logical operators: and / or

'and' (also '&&') is true only when both sides are true. 'or' (also '||') is true when at least one side is true. Both skip the right side when the left side already determines the result. For example, 'false and X' is always false, so X is never evaluated. This matters when the right side could produce null or has a side effect.

```
  assert(1 > 0 and 2 > 1, "Both conditions true");
  assert(1 > 2 or 3 > 2, "Second condition true");
  assert!(1 > 2 && 3 > 2), "First false → whole 'and' is false");
  assert(!(1 > 2 || 3 > 4), "Both false → 'or' is false");
```

8.0.2. Null in boolean context

Division by zero (and other failed operations) produce null. Null in a boolean context is treated as false, so '!' is true. This lets you write guard clauses without a separate null-check syntax:

```
if !result { ... handle missing value ... }
```

```
  zero = 0;
  assert(!(12 / zero), "null from division-by-zero is false-like");
  assert(12 > 0, "positive integer is true");
```

8.0.3. Bitwise operators

While 'and'/'or' work on true/false values, bitwise operators work on the individual bits of an integer. They are useful for flags, masks, and low-level data manipulation.

```
'&'  — keeps only bits set in BOTH operands (AND)
'|'  — keeps bits set in EITHER operand (OR)
'<<' — shift bits left N positions  (×2^N)
'>>' — shift bits right N positions (÷2^N)
```

Tip: ‘&’ binds less tightly than comparison operators. Use parentheses when you mix bitwise and comparison expressions in the same condition.

```
assert((0x0f & 0xa8) == 8, "Bitwise AND: keeps only bits in both");
assert((0xf0 | 0x0f) == 0xff, "Bitwise OR: combines bits from either");
assert(1 << 4 == 16, "Left shift: 1 × 2^4 = 16");
assert(256 >> 3 == 32, "Right shift: 256 ÷ 2^3 = 32");
```

8.0.4. Negation

‘!’ is the logical NOT operator. It flips any boolean expression.

```
assert(!false, "not false is true");
assert!(1 > 2), "not false comparison is true");
```

8.0.5. Formatting booleans

Booleans can be embedded in a format string like any other value. The ‘^’ alignment specifier centres the value in a field of given width. ‘<’ aligns left, ‘>’ aligns right (right-alignment is the default).

```
assert("1{true:^7}2" == "1 true  2", "Centred boolean in field of width 7");
assert("{false}" == "false", "Plain boolean format");
```

8.0.6. ‘&’ vs ‘and’

‘&’ is bitwise AND on integers; ‘and’ (or ‘&&’) is logical AND on booleans. Writing ‘a & b’ where both sides are booleans is a compile error — the compiler requires ‘and’ or ‘&&’ for boolean logic.

```
flag_a = 3 > 1;
```

true

```
flag_b = 4 > 2;
```

true

```
assert(flag_a and flag_b, "Correct: logical AND on two booleans");
}
```

9. Float

A ‘float’ stores a number with a decimal point, like 3.14 or −0.001. It gives you about 15 significant digits of precision — enough for scientific work, games, and most real-world maths. When you write a number with a decimal point in Loft, it is automatically a float.

```
fn main() {
```

Write a decimal point in a literal and Loft treats the whole expression as a float. The usual arithmetic operators (+, -, *, /) all work the way you would expect.

```
  assert(1.5 + 0.5 == 2.0, "Float addition");
  assert(3.0 / 2.0 == 1.5, "Float division");
  assert(2.0 * 1.5 == 3.0, "Float multiplication");
```

The ‘f’ suffix selects single-precision floats, which take half the memory of a regular float but have fewer decimal digits of accuracy. This trade-off is common in graphics and audio code where exact decimal values matter less than speed or memory.

```
  x = 0.1f + 2 * 1.0f;
  assert(x == 2.1f, "Single precision float");
```

‘as float’ converts an integer to a float so you can mix them in calculations. Putting a float inside ‘{...}’ converts it to text for display. You can also go the other way: ‘“1.5” as float’ parses the text back to a number.

```
  assert(3 as float == 3.0, "Integer to float");
  assert("1.5" as float == 1.5, "Text to float");
  assert("{1.5}" == "1.5", "Float formatted as text");
```

9.0.1. Formatting Floats

Put a colon after the value inside ‘{...}’ to control how it looks. ‘{value:width.precision}’ — ‘width’ is the minimum number of characters printed (padded with spaces on the left); ‘precision’ fixes the number of decimal places. You can use either part on its own: ‘{value:.2}’ just fixes decimal places, ‘{value:5}’ just sets the minimum width.

```
  assert("{1.2:4.2}" == "1.20", "Float with width and precision");
  assert("{334.1:.2}" == "334.10", "Float with precision only");
  assert("{1.4:5}" == "  1.4", "Float with width only");
```

9.0.2. Math Functions

‘PI’ is a built-in constant (approximately 3.14159265358979). Multiplying it by 1000 and rounding gives 3142, which confirms the value is correct.

```
  assert(round(PI * 1000.0) == 3142.0, "PI constant");
```

‘pow(base, exponent)’ raises a number to a power — this is exponentiation. Note: ‘^’ in Loft is bitwise XOR, NOT exponentiation. Always use pow() for powers. ‘log(value, base)’ is the inverse: log(x, b) answers “b to the what power equals x?” Here: 4^5 = 1024, and log(1024, 2) = 10 because 2^10 = 1024.

```
  assert(log(pow(4.0, 5), 2) == 10.0, "log base 2 of 4^5");
```

‘sin’ and ‘cos’ work in radians, not degrees. A full circle is 2*PI radians; a half circle (180 degrees) is PI radians. sin(PI) should be exactly zero but computers use approximations for decimal numbers, so you may get something like 0.0000000001 instead. We use ceil() to round it up to 0. cos(PI) is exactly −1, so the whole expression ceil(0 + −1 * 1000) lands at −1000.

```
  assert(ceil(sin(PI) + cos(PI) * 1000) == -1000.0, "sin and cos");
```

‘abs()’ returns the absolute value — the distance from zero, always non-negative. Useful any time you care about magnitude but not direction.

```
  assert(abs(-2.5) == 2.5, "Absolute value of float");
```

‘round()’ picks the nearest whole number (0.5 rounds up). ‘ceil()’ always rounds up to the next whole number, even for 2.01. ‘floor()’ always rounds down to the previous whole number, even for 2.99. All three give back a float, not an integer — so 3.0, not 3.

```
  assert(round(2.6) == 3.0, "round up");
  assert(round(2.4) == 2.0, "round down");
  assert(ceil(2.1) == 3.0, "ceil");
  assert(floor(2.9) == 2.0, "floor");
```

Single-precision floats work inside format strings just like regular floats.

```
  assert("a{0.1f + 2 * 1.0f}b" == "a2.1b", "Single-precision format");
}
```

10. Functions

Functions let you give a reusable piece of logic a name so you can call it from multiple places without copying code. This page covers: declaring functions, default argument values, reference parameters (so a function can modify the caller's variable), early return, const parameters, and type-based dispatch.

10.0.1. Declaring Functions

The keyword 'fn' starts a function definition. List parameters as 'name: type' separated by commas. '-> type' declares what the function gives back. The last expression in the body is returned automatically — no 'return' needed there. Default values let callers skip optional arguments.

```
fn greet(name: text, greeting: text = "Hello") -> text {
  greeting + ", " + name + "!"
}
```

10.0.2. Default arguments that involve other arguments

Loft defaults must be literal values, not expressions referencing other parameters. When one argument's default should be derived from another, use a sentinel value and compute the actual default inside the function body. The convention is to use −1 (or 0) as "not provided" for integer parameters.

```
fn substring(t: text, start: integer = 0, end: integer = -1) -> text {
  actual_end = if end < 0 { len(t) } else { end };
  t[start..actual_end]
}
```

10.0.3. Named Arguments

When a function has several parameters with defaults, you can skip middle ones by naming the arguments you want to provide. Positional arguments come first; once you use a name, all remaining arguments must also be named.

```
fn connect(host: text, port: integer = 8080, tls: boolean = true) -> text {
  "{host}:{port}:{tls}"
}
```

10.0.4. Reference Parameters and Defaults

A function with no '-> type' is called for its side effects, not its result. Adding '&' before a parameter type makes it a reference: the function receives a direct link to the caller's variable and can read or write it. Without '&', Loft copies the value in, so changes inside the function stay local. Default values (written '= value' after the type) are substituted when the caller omits that argument.

```
fn add(a:  & integer, b: integer, c: integer = 0) {
  a += b + c;
}
```

10.0.5. Early Return

Use 'return value;' to exit a function before reaching the end. This is handy for guard clauses: handle the special cases first, then write the normal path without extra nesting.

```
fn classify(n: integer) -> text {
  if n < 0 {
    return "negative";
  }
  if n == 0 {
    return "zero";
  }
  "positive"
}
```

10.0.6. Const and Reference Parameters

'const' on a parameter tells the compiler "this function must never change this value." The compiler enforces it — any assignment to a const parameter is a compile error. '&' is the opposite promise: "this function will modify this value." Declaring '&' without actually writing to the parameter is also a compile error. These rules make it easy to read a function signature and know what it does to its inputs.

```
fn scale(a: integer, factor: integer) -> integer {
  a * factor
}
```

10.0.7. Type-Based Dispatch

You can define two functions with the same name as long as their parameter types differ. Loft picks the right one at compile time based on the type of the argument you pass.

```
fn describe_int(v: integer) -> text {
  "integer:{v}"
}
```

```
fn describe_text(v: text) -> text {
  "text:{v}"
}
```

10.0.8. Function References

'fn <name>' creates a reference to a named function that you can store or pass around. The compiler checks that the name exists and is a function — a typo is a compile error. The result has type 'fn(param_types) -> return_type' and can be:

```
- stored in a variable,
- called directly with 'f(args)', and
- passed as a parameter to another function.
```

```
fn double_it(x: integer) -> integer {
  x * 2
}
```

```
fn negate_it(x: integer) -> integer {
  - x
}
```

```
fn apply_fn(f:  fn(integer) -> integer, x: integer) -> integer {
  f(x)
}
```

10.0.9. Lambda Expressions

A lambda is an anonymous (unnamed) function written right where you need it. Use the full form when you want to be explicit about types:

```
fn(param: type) -> return_type { body }
```

Use the short form (pipe-bar syntax) when the call site already knows the types:

```
|param| { body }
```

Both forms work anywhere a function reference is expected — especially with map, filter, and reduce. The short form does not allow type annotations; use the full fn(...) form when you need explicit types. Lambdas that need outer variables can capture them: see the Closures page.

```
fn main() {
```

Passing both arguments explicitly overrides the default.

```
  assert(greet("World", "Hi") == "Hi, World!", "Explicit argument");
```

Leaving out the second argument causes Loft to use the default “Hello”.

```
  assert(greet("World") == "Hello, World!", "Default argument");
```

‘add’ takes ‘a’ by reference, so every call updates the original variable ‘v’. Watch how v accumulates: 1 -> 3 -> 8.

```
  v = 1;
  add(v, 2);
```

v = 1 + 2 = 3

```
  add(v, 4, 1);
```

v = 3 + 4 + 1 = 8

```
assert(v == 8, "Reference parameter: {v}");
```

‘classify’ uses early returns to handle each case separately.

```
assert(classify(-5) == "negative", "Classify negative");
assert(classify(0) == "zero", "Classify zero");
assert(classify(3) == "positive", "Classify positive");
```

‘scale’ marks both parameters ‘const’, so the compiler verifies they are not changed.

```
assert(scale(3, 7) == 21, "scale(3,7)");
```

Loft selects the right function based on the type of the argument.

```
assert(describe_int(42) == "integer:42", "Integer describe");
assert(describe_text("hi") == "text:hi", "Text describe");
```

A function reference is stored in a variable with type ‘fn(...) -> ...’. Calling it looks exactly like a regular function call.

```
f = double_it;
assert(f(5) == 10, "fn-ref stored and called: {f(5)}");
```

Pass function references as arguments to higher-order functions.

```
assert(apply_fn(double_it, 7) == 14, "fn-ref as arg (double)");
assert(apply_fn(negate_it, 3) == -3, "fn-ref as arg (negate)");
```

Lambdas with map, filter, and reduce. Use |...| short form — types are inferred from the call site.

```
nums = [1, 2, 3, 4, 5];
doubled = map(nums, |x| { x * 2 });
assert(doubled[0] == 2, "lambda map first element");
assert(doubled[4] == 10, "lambda map last element");
```

filter keeps only elements where the lambda returns true.

```
evens = filter(nums, |x| { x % 2 == 0 });
assert(evens[0] == 2, "filter evens first");
assert(evens[1] == 4, "filter evens second");
```

reduce collapses the whole list into one value. The lambda receives the running total (acc) and the next element (x).

```
total = reduce(nums, 0, |acc, x| { acc + x });
assert(total == 15, "reduce sum: {total}");
```

Use fn(...) long form when you need explicit types — for example when storing a lambda in a variable or passing it without call-site context.

```
  negate = fn(x: integer) -> integer { -x };
  assert(negate(7) == -7, "stored lambda: {negate(7)}");
  assert(apply_fn(|x| { x * x }, 6) == 36, "lambda as arg: 6^2");
```

— Named arguments —

```
  assert(connect("example.com") == "example.com:8080:true", "all defaults");
  assert(connect("example.com", tls: false) == "example.com:8080:false", "skip
port");
  assert(connect(host: "example.com", tls: false) == "example.com:8080:false",
"all named");
  assert(connect("example.com", port: 443, tls: false) ==
"example.com:443:false", "mixed");
  assert(greet(name: "World") == "Hello, World!", "named with default greeting");
```

— Default argument that depends on another argument — 'end' defaults to −1 (sentinel); inside the body it becomes len(t).

```
  assert(substring("hello", 1, 3) == "el", "explicit start and end");
  assert(substring("hello", 2) == "llo", "end defaults to len(t)");
  assert(substring("hello") == "hello", "both defaults");
}
```

11. Vector

A vector is an ordered list of values that can grow and shrink while your program runs. Every element must have the same type — you cannot mix integers and text in one vector. Write a vector literal with square brackets: '[1, 2, 3]'. Loft automatically manages the storage, so vectors grow as you add elements without you having to say how large they will be up front.

11.0.1. Transforming vectors: map, filter, reduce

'map', 'filter', and 'reduce' each take a function and apply it to the vector. Pass the function using 'fn <name>' to refer to a named function by name.

```
map(v, f)          — apply f to every element; returns a new vector
filter(v, pred)    — keep only elements for which pred returns true
reduce(v, f, init) — combine all elements into a single value
```

```
fn triple(x: integer) -> integer {
  x * 3
}
```

```
fn is_even_n(x: integer) -> boolean {
  x % 2 == 0
}
```

```
fn sum_acc(acc: integer, x: integer) -> integer {
  acc + x
}
```

```
fn mul_acc(acc: integer, x: integer) -> integer {
  acc * x
}
```

```
fn main() {
```

Create a vector with a literal and loop over it with 'for'. Two special loop annotations help when you need to know where you are:

```
'v#first'  is true only on the very first iteration — useful for skipping
separators.
'v#index'  holds the zero-based position of the current element.
```

```
  x =[1, 3, 6, 9];
  b = "";
  for v in x {
    if !v#first {
      b += " ";
    }
```

```
    b += "{v#index}:{v}"
  }
  assert(b == "0:1 1:3 2:6 3:9", "result {b}");
```

'+=' appends another vector to the end of an existing one. You can filter and delete elements in a single pass:

```
Add 'if condition' after 'in vector' to visit only matching elements.
Write 'v#remove' inside the loop body to delete the current element.
```

Here we keep only multiples of 3 by removing everything else. Note: you cannot append to a vector (v += [...]) while iterating over it — that is a compile error because the loop would then visit the new elements too, which could loop forever.

```
  x +=[12, 14, 15];
  for v in x if v % 3 != 0 {
    v#remove;
  }
  assert("{x}" == "[3,6,9,12,15]", "result {x}");
```

11.0.2. Clearing

'v.clear()' removes all elements, setting the length to 0. The underlying storage is kept so appending afterwards is efficient.

```
  x.clear();
  assert(x.len() == 0, "clear empties the vector");
  x += [99];
  assert(x[0] == 99, "append after clear works");
```

11.0.3. Slicing

A slice gives you a window into part of a vector without copying it. 'v[a..b]' contains elements at positions a, a+1, ..., b-1 (b is excluded). 'v[a..]' goes from position a to the very last element. 'v[..b]' goes from the beginning up to (but not including) position b.

```
  pows =[1, 2, 4, 8, 16];
  assert("{pows[1..3]}" == "[2,4]", "Sub-vector");
  assert("{pows[3..]}" == "[8,16]", "Open-ended sub-vector");
  assert("{pows[..3]}" == "[1,2,4]", "Open-start sub-vector");
```

Accessing an index that does not exist is safe: Loft returns null instead of crashing. You can check for null with '!' (logical not) because null is falsy.

```
  assert(!pows[10], "Out-of-bounds access returns null");
```

11.0.4. Comprehensions

A comprehension is a concise way to build a new vector from a formula. Syntax: '[for variable in range { expression }]' Add 'if condition' to include only elements that satisfy the condition. This is much shorter than creating an empty vector and appending in a loop.

```
evens =[for n in 1..10 if n % 2 == 0 {
  n
}];
assert("{evens}" == "[2,4,6,8]", "Filtered comprehension");
doubled =[for n in 1..6 {
  n * 2
}];
assert("{doubled}" == "[2,4,6,8,10]", "Doubled comprehension");
```

Embedding a for loop directly inside a format string '{...}' produces a formatted list. The format specifier after ':' is applied to every element in the result. ':02' means "at least 2 digits wide, padded with zeros on the left".

```
  assert("{for n in 1..7 {n*2}:02}" == "[02,04,06,08,10,12]", "Formatted vector
loop");
```

11.0.5. Reverse Iteration

Wrap a range in 'rev()' to step through elements from the last index to the first. 'rev(0..=3)' covers indices 0, 1, 2, 3 — the '=' makes the upper end inclusive. Here we visit pows[3]=8, pows[2]=4, pows[1]=2, pows[0]=1, building 8421 digit by digit.

```
c = 0;
for e in pows[rev(0..=3)] {
  c = c * 10 + e;
}
assert(c == 8421, "Reverse sub-vector iteration");
```

You can fill a vector with many copies of the same value using '; count' syntax:

```
[SomeStruct { field: value }; 16]
```

This creates 16 identical copies in one expression. See 08-struct.loft for examples.

11.0.6. Passing vectors to functions

When you pass a vector to a function, the function receives a **slice** — a start position and a length inside the storage of the caller. This is efficient because no data is copied, but it has an important consequence: the function can read and modify existing elements (because it shares the same storage), but it cannot grow or shrink the vector. Appending with '+=' inside the function creates a local copy that the caller never sees.

To let a function append to the caller's vector, mark the parameter with '&'. This tells the compiler to propagate structural changes (appends, clears) back to the caller when the function returns.

```
fn append_one(v: &vector<integer>, x: integer) { v += [x]; }
```

Without '&', only element-level mutations are visible to the caller:

```
fn set_first(v: vector<integer>, x: integer) { v[0] = x; }  // caller sees the
change
fn try_push(v: vector<integer>, x: integer) { v += [x]; }   // caller does NOT
see the append
```

The same rule applies to slices: 'v[2..5]' passed to a function is a narrower window into the same storage, so element writes are visible but appends are not.

11.0.7. Higher-order functions

'map' applies a function to every element and returns a new vector.

```
  nums =[1, 2, 3, 4, 5];
  tripled = map(nums,  triple);
  assert("{tripled}" == "[3,6,9,12,15]", "map triple: {tripled}");
```

'filter' keeps only elements for which the predicate returns true.

```
  evens2 = filter(nums,  is_even_n);
  assert("{evens2}" == "[2,4]", "filter evens: {evens2}");
```

'reduce' folds all elements into a single value, starting from an initial accumulator. Argument order: reduce(vector, initial_value, combiner).

```
  total = reduce(nums, 0,  sum_acc);
  assert(total == 15, "reduce sum: {total}");
  product = reduce(nums, 1,  mul_acc);
  assert(product == 120, "reduce product: {product}");
```

You can chain these calls: filter first, then map, then reduce.

```
  result = reduce(map(filter(nums,  is_even_n),  triple), 0,  sum_acc);
  assert(result == 18, "filter+map+reduce: {result}");
}
```

12. Structs

A struct groups related values under one name. Instead of keeping a product's name, price, and stock count in three separate variables that can drift out of sync, you bundle them together into a Product struct. Each piece of data inside the struct is called a field. You read and write fields using dot notation: 'product.price'. Here is a simple product record. All four fields are plain integers or text.

```
struct Product {
  name: text,
  price: integer,
  stock: integer
}
```

12.0.1. Field Constraints

You can restrict what values a field may hold. 'limit(min, max)' rejects any value outside that range at runtime. 'not null' tells the field that zero is a real data value — without it, zero is treated as "no value" (null). Colour channels can all be zero (pure black is a valid colour), so all three need 'not null'. Fields you omit in a constructor receive zero (or null for nullable fields) by default.

```
struct Colour {
  r: integer limit(0, 255) not null,
  g: integer limit(0, 255) not null,
  b: integer limit(0, 255) not null
}
```

12.0.2. Methods

A method is a function whose first parameter is named 'self'. Loft uses the type of 'self' to decide which struct the method belongs to. Call it with dot notation: 'c.to_hex()'. A method can read fields via 'self.field' and return any type.

```
fn to_hex(self: Colour) -> integer {
  self.r * 0x10000 + self.g * 0x100 + self.b
}
```

A method can also return a completely new struct value. Write '-> StructName' as the return type and construct the value in the body. The original struct is not modified — the caller gets a brand-new copy.

```
fn dimmed(self: Colour) -> Colour {
  Colour {r: self.r / 2, g: self.g / 2, b: self.b / 2 }
}
```

12.0.3. Defaults and Computed Fields

Write '= expression' after the type to set a stored default, filled at construction. Use 'computed(expression)' for a field that recalculates every time you read it — it takes no space in the record and always reflects the current state. Inside both, '$' refers to the struct.

```
struct Item {
  name: text,
  name_length: integer = len($.name)
}
```

```
struct Circle {
  radius: float,
  area: float computed(3.14159265 * $.radius * $.radius)
}
```

12.0.4. Storing many structs in a vector

'Area' uses smaller integer types to save memory: 'u16' holds values 0–65535, and 'u8' holds values 0–255. This matters when you have millions of records (such as tiles in a game map) and memory usage counts.

```
struct Area {
  height: u16,
  terrain: u8,
  water: u8,
  direction: u8
}
```

```
fn main() {
```

Constructing a struct: name the fields you want to set. Fields you leave out default to zero (or null for nullable fields).

```
  apple = Product {name: "Apple", price: 120, stock: 50 };
  assert(apple.price == 120, "price field: {apple.price}");
  assert(apple.name == "Apple", "name field: {apple.name}");
```

You can read and write individual fields after construction.

```
  apple.stock -= 1;
  assert(apple.stock == 49, "stock after one sale: {apple.stock}");
```

A field omitted from the constructor gets zero as its default.

```
  col = Colour {r: 128, b: 128 };
  assert(col.g == 0, "omitted green channel defaults to zero");
```

Formatting a struct shows all fields compactly.

```
  assert("{col}" == "{{r:128,g:0,b:128}}", "Struct compact formatting");
```

':j' produces JSON output — a text format used by many web APIs and config tools.

```
  assert("{col:j}" == "{{\"r\":128,\"g\":0,\"b\":128}}", "JSON format");
```

Call a method on a variable using dot notation.

```
  purple = Colour {r: 128, b: 128 };
  assert("{purple.to_hex():x}" == "800080", "hex method result");
```

You can call a method directly on a constructor expression.

```
  assert(Colour {r: 255, g: 0, b: 0 }.to_hex() == 0xff0000, "method on
constructor");
```

'dimmed' returns a new Colour with each channel halved. The original variable is unchanged.

```
  dark = purple.dimmed();
  assert(dark.r == 64, "dimmed r: {dark.r}");
  assert(dark.b == 64, "dimmed b: {dark.b}");
  assert(dark.g == 0, "dimmed g: {dark.g}");
```

Stored defaults are filled once at construction time.

```
  it = Item {name: "hello" };
  assert(it.name_length == 5, "stored default name_length: {it.name_length}");
```

computed() fields recalculate on every access — changing radius updates area.

```
  ci = Circle { radius: 5.0 };
  assert(ci.area > 78.0, "computed area: {ci.area}");
  ci.radius = 10.0;
  assert(ci.area > 314.0, "recomputed area: {ci.area}");
```

Fill a vector with copies of the same struct using '; count' syntax. This creates 16 Area records, all with the same initial values.

```
  map =[Area {height: 0, terrain: 1, water: 1, direction: 1 };
  16];
  assert("{map[3]}" == "{{height:0,terrain:1,water:1,direction:1}}", "tile
record");
  map[3].height = 200;
  assert(map[3].height == 200, "individual tile update");
```

12.0.5. sizeof

'sizeof(Type)' returns the packed byte size used when the type is stored as a struct field or vector element. Range-constrained integer types like u8 and u16 report their packed size, not the 4-byte stack slot size.

```
  assert(sizeof(integer) == 4, "integer: 4 bytes");
  assert(sizeof(u8) == 1, "u8: 1 byte (packed)");
  assert(sizeof(u16) == 2, "u16: 2 bytes (packed)");
  assert(sizeof(Colour) == 3, "Colour: 3 × u8 = 3 bytes");
  assert(sizeof(Area) == 5, "Area: u16 + 3 × u8 = 5 bytes");
}
```

13. Enums

An enum defines a fixed set of named choices. Instead of using raw numbers like 0 = north, 1 = east — which nobody can read six months later — you give each option a name. The compiler then checks every comparison and conversion, so a typo becomes a compile error instead of a silent wrong answer. Plain enums are just names that can be compared, ordered, and converted to and from text. Their order matches their declaration order.

```
enum Direction {
  North,
  East,
  South,
  West
}
```

13.0.1. Enums that carry data

Sometimes different choices have different shapes of data. A 'Circle' needs one number (radius); a 'Rect' needs two (width and height). Struct enums let each variant carry its own fields, keeping all the shape kinds in one type while still letting you work with each variant naturally.

```
enum Shape {
  Circle {radius: float },
  Rect {width: float, height: float }
}
```

13.0.2. Methods that work on any variant

Write the same function name for each variant, each taking 'self' of that variant's type. Loft automatically picks the right version at runtime based on which variant is actually stored in the variable. This lets you call shape.area() without caring which kind of shape it is.

```
fn area(self: Circle) -> float {
  PI * pow(self.radius, 2)
}
```

```
fn area(self: Rect) -> float {
  self.width * self.height
}
```

Polymorphic methods can return text just as well as numbers. Each variant can produce its own human-readable description.

```
fn describe(self: Circle) -> text {
  "circle with radius {self.radius}"
}
```

```
fn describe(self: Rect) -> text {
  "rect {self.width} by {self.height}"
}
```

13.0.3. Enum methods

Plain enum variants can also have methods. The 'self' parameter carries the current direction value, and the method can return any type. Here 'opposite()' flips North↔South and East↔West.

```
fn opposite(self: Direction) -> Direction {
  if self == North {
    South
  } else if self == South {
    North
  } else if self == East {
    West
  } else {
    East
  }
}
```

```
fn main() {
```

Plain enum values are ordered by declaration position. North comes first (smallest), West comes last (greatest).

```
  d = East;
  assert(d == East, "Enum equality");
  assert(d != North, "Enum inequality");
  assert(d > North, "East declared after North, so it is greater");
  assert(West > South, "West declared last, so it is greatest");
```

Convert between text and enum in both directions.

```
  assert("{d}" == "East", "Format plain enum as text");
  assert("West" as Direction == West, "Parse text to enum");
```

Use a plain enum method like any other method.

```
  assert(d.opposite() == West, "opposite of East is West");
  assert(North.opposite() == South, "opposite of North is South");
```

Struct enum variants are constructed exactly like regular structs.

```
  c = Circle {radius: 1.0 };
  assert(c.radius == 1.0, "Circle radius field");
  r = Rect {width: 4.0, height: 5.0 };
  assert(r.width * r.height == 20.0, "Rect manual area");
```

Calling area() on a Circle runs the Circle version; on a Rect it runs the Rect version — chosen automatically at runtime.

```
  assert(round(c.area() * 1000) == round(PI * 1000), "Circle area = PI*r^2");
  assert(r.area() == 20.0, "Rect area = width*height");
```

describe() uses format strings with field access inside each variant's method.

```
  assert(c.describe() == "circle with radius 1", "describe circle:
{c.describe()}");
  assert(r.describe() == "rect 4 by 5", "describe rect: {r.describe()}");
```

13.0.4. Stubs for missing implementations

If a variant intentionally has no implementation of a method, the compiler emits a warning. Provide an empty-body stub to silence it:

```
fn area(self: SomeVariant) -> float { }
```

A stub returns null at runtime and suppresses the warning.

13.0.5. Match expressions on enums

Match picks a code path based on the active variant. You must handle every variant, or include a _ wildcard arm that catches the rest.

```
  axis = match d {
    North | South => "vertical",
    East | West => "horizontal"
  };
  assert(axis == "horizontal", "or-pattern: East matches East|West");
```

When a variant has fields, name them inside braces to use them in the arm body.

```
  label = match c {
    Circle { radius } => "r={radius}",
    Rect { width, height } => "{width}x{height}"
  };
  assert(label == "r=1", "field destructuring in match arm");
```

13.0.6. Guard clauses

An arm can have an `if` guard after the pattern. If the guard fails, matching falls through to the next arm. Because the guard can fail, a guarded arm alone does not prove the variant is handled — you still need a wildcard _ or an unguarded arm for that variant.

```
  area = match c {
    Circle { radius } if radius > 0.0 => PI * radius * radius,
    _ => 0.0
```

```
  };
  assert(round(area * 1000) == round(PI * 1000), "guarded match on Circle");
```

13.0.7. Scalar match

Match also works on integers, text, floats, booleans, and characters. Arms can be literals, ranges, `null`, or _.

```
  grade = match 85 {
    90..=100 => "A",
    80..90   => "B",
    _        => "C"
  };
  assert(grade == "B", "range pattern 80..90 matches 85");
```

Or-patterns work on scalars too.

```
  kind = match 2 {
    1 | 2 | 3 => "low",
    _         => "high"
  };
  assert(kind == "low", "scalar or-pattern");
```

A `null` pattern matches when the value is absent (e.g. division by zero).

```
  zero = 0;
  check = match 1 / zero {
    null => "absent",
    _    => "present"
  };
  assert(check == "absent", "null pattern matches div-by-zero");
```

Character literals work in match arms.

```
  vowel = match 'e' {
    'a' | 'e' | 'i' | 'o' | 'u' => true,
    _ => false
  };
  assert(vowel, "character or-pattern");
}
```

14. Sorted

A ‘sorted’ collection holds records in order by one or more fields you choose as the sort criteria (called “key fields”). You can look up a record by those fields instantly and iterate all records in that order. The collection stays sorted as you add new elements — no manual sorting needed. Declare the key fields inside angle brackets: ‘field’ sorts A⟶Z / 0⟶9 (ascending), ‘-field’ sorts Z⟶A / 9⟶0 (descending).

```
struct Elm {
  key: text,
  value: integer
}
```

```
struct Db {
  map: sorted < Elm[-key] >
}
```

```
fn main() {
```

14.0.1. Adding Elements

You initialise a sorted collection the same way as a vector. The elements are automatically placed in the right position as you add them.

```
  db = Db {map: [
  Elm {key: "One", value: 1 },
  Elm {key: "Two", value: 2 },
  Elm {key: "Three", value: 3 },
  Elm {key: "Four", value: 4 }
  ] };
```

14.0.2. Looking Up by Key

Use square brackets with the key value to find a record instantly. If the key is not present the result is null — you can check it with ‘!’.

```
  assert(db.map["Two"].value == 2, "Key lookup: Two");
  assert(db.map["Three"].value == 3, "Key lookup: Three");
  assert(!db.map["Five"], "Missing key returns null");
  assert(!db.map[null], "Null key returns null");
```

Append new elements with ‘+=’; they are placed in the correct sorted position.

```
  db.map +=[Elm {key: "Zero", value: 0 }];
  assert(db.map["Zero"].value == 0, "Newly added element");
```

14.0.3. Iterating in Order

A ‘for’ loop over a sorted collection visits elements in key order. Here the key is ‘-key’ (descending text), so the order is: Zero, Two, Three, One, Four.

```
sum = 0;
for v in db.map {
  sum = sum * 10 + v.value;
}
```

Zero(0), Two(2), Three(3), One(1), Four(4) ⟶ 0*10+2=2, *10+3=23, *10+1=231, *10+4=2314

```
assert(sum == 2314, "Sorted iteration total: {sum}");
```

14.0.4. Iterating in Reverse

Wrap the collection in rev() to visit elements from last to first key order. Here the collection is sorted by '-key' (descending text), so the stored order is Zero, Two, Three, One, Four. rev() visits them in the opposite order.

```
rev_sum = 0;
for v in rev(db.map) {
  rev_sum = rev_sum * 10 + v.value;
}
```

Four(4), One(1), Three(3), Two(2), Zero(0) ⟶ 4*10+1=41, *10+3=413, *10+2=4132, *10+0=41320

```
assert(rev_sum == 41320, "Reverse iteration total: {rev_sum}");
```

14.0.5. Loop Helpers: #first, #count, and #remove

'v#first' is true for the very first element visited. 'v#count' is a running index starting at 0. 'v#remove' removes the current element while iterating.

```
first_key = "";
labels = "";
for v in db.map {
  if v#first {
    first_key = v.key
  }
  if !v#first {
    labels += ","
  }
  labels += "{v#count}:{v.key}"
}
assert(first_key == "Zero", "First element: {first_key}");
assert(labels == "0:Zero,1:Two,2:Three,3:One,4:Four", "Labels: {labels}");
```

Remove elements with value > 2 while iterating.

```
for v in db.map if v.value > 2 {
  v#remove
}
sum2 = 0;
```

```
  for v in db.map {
    sum2 += v.value
  }
```

Remaining: Zero(0), Two(2), One(1) ⟶ sum = 3

```
  assert(sum2 == 3, "Sum after remove: {sum2}");
```

14.0.6. Removing by Key

Assigning null to a sorted subscript removes the element with that key. Removing a key that is not present is a safe no-op.

```
  db2 = Db {map: [
  Elm {key: "A", value: 10 },
  Elm {key: "B", value: 20 },
  Elm {key: "C", value: 30 }
  ] };
  db2.map["B"] = null;
  assert(!db2.map["B"], "B was removed");
  assert(db2.map["A"].value == 10, "A still present");
  assert(db2.map["C"].value == 30, "C still present");
  db2.map["missing"] = null;
```

no-op; does not panic

```
  sum3 = 0;
  for v in db2.map {
    sum3 += v.value
  }
  assert(sum3 == 40, "Sum after key removal: {sum3}");
```

Note: #index is not available on sorted — use #count for a sequential counter.

```
}
```

```
struct Word {
  name: text,
  score: integer
}
```

```
struct Dict {
  words: sorted<Word[name]>
}
```

```
fn main_ranges() {
```

14.0.7. Iterating a Key Range

Use an `if` guard inside a `for` loop to visit only records whose key falls within a range. Because the collection is sorted the guard still sees all elements, but the body only runs for the matching ones.

```
d = Dict { words: [
  Word { name: "apple",      score: 5 },
  Word { name: "banana",     score: 3 },
  Word { name: "cherry",     score: 8 },
  Word { name: "date",       score: 2 },
  Word { name: "elderberry", score: 1 }
] };
```

Closed range: keys >= “banana” and <= “date”.

```
range_total = 0;
for w in d.words if w.name >= "banana" && w.name <= "date" {
  range_total += w.score
}
assert(range_total == 13, "banana(3)+cherry(8)+date(2) = {range_total}");
```

14.0.8. Open-ended Range (tail)

All records from “cherry” onward to the end of the sorted order.

```
tail_count = 0;
for w in d.words if w.name >= "cherry" {
  tail_count += 1
}
assert(tail_count == 3, "cherry, date, elderberry = {tail_count}");
```

14.0.9. Open-ended Range (head)

All records up to and including “banana”.

```
head_names = "";
for w in d.words if w.name <= "banana" {
  head_names += w.name + " "
}
assert(head_names == "apple banana ", "apple and banana: '{head_names}'");
```

14.0.10. Range outside the for (building a result vector)

Collect matching records into a vector for later use.

```
  subset = [for w in d.words if w.name >= "banana" && w.name <=
"cherry" { w.score }];
  assert(subset[0] == 3, "first in range: {subset[0]}");
  assert(subset[1] == 8, "second in range: {subset[1]}");
}
```

filling and finding values

15. Index

An 'index' lets you find records instantly by key and iterate over ranges of keys in order. It supports multi-part keys: you can sort by a primary key and break ties with a secondary key. Declare the key fields inside angle brackets: 'field' sorts ascending, '-field' descending. Note: each record can only belong to one index at a time.

```
struct Elm {
  nr: integer,
  key: text,
  value: integer
}
```

```
struct Db {
  map: index < Elm[nr, -key] >
}
```

```
fn main() {
```

15.0.1. Adding and Looking Up Records

Fill the index just like a vector. Elements are automatically kept in key order. This index is sorted first by 'nr' (ascending), then by 'key' (descending) for ties.

```
  db = Db {map: [
  Elm {nr: 101, key: "One", value: 1 },
  Elm {nr: 92, key: "Two", value: 2 },
  Elm {nr: 83, key: "Three", value: 3 },
  Elm {nr: 83, key: "Four", value: 4 },
  Elm {nr: 83, key: "Five", value: 5 },
  Elm {nr: 63, key: "Six", value: 6 }
  ] };
```

Provide all key fields together inside brackets to find a record. Supply fewer fields to match all records that share a prefix.

```
  assert(db.map[101, "One"].value == 1, "Key lookup");
  assert(!db.map[12, ""], "Missing key returns null");
  assert(!db.map[83, "One"], "Wrong secondary key returns null");
```

15.0.2. Iterating in Order

A 'for' loop visits all elements in key order.

```
  total = 0;
  for r in db.map {
    total += r.value;
  }
  assert(total == 21, "Sum of all values: {total}");
```

15.0.3. Range Queries

Provide a range in the first key position to visit only the matching slice. '[83..92, "Two"]' means: nr from 83 up to (not including) 92, and key from "Two" down (since the key is sorted descending, "Two" is the start of that descending segment).

```
sum = 0;
for v in db.map[83..92, "Two"] {
  sum = sum * 10 + v.value;
}
```

Elements matched: (83, Three, 3), (83, Four, 4), (83, Five, 5)

```
assert(sum == 345, "Range iteration result: {sum}");
```

15.0.4. Loop Helpers: #first and #count

'r#first' is true for the very first element visited. 'r#count' is a running counter starting at 0. Note: #index is not meaningful on index collections — use #count instead.

```
first_nr = 0;
count_total = 0;
for r in db.map {
  if r#first {
    first_nr = r.nr
  }
  count_total = r#count
}
assert(first_nr == 63, "First element by key: {first_nr}");
assert(count_total == 5, "Total elements: {count_total}");
```

15.0.5. Removing by Key

Assigning null to an index subscript removes the element with that exact key combination. Removing a key that is not present is a safe no-op.

```
db.map[92, "Two"] = null;
assert(!db.map[92, "Two"], "element (92, Two) was removed");
assert(db.map[101, "One"].value == 1, "element (101, One) still present");
db.map[999, "Z"] = null;
```

no-op; does not panic

```
total2 = 0;
for r in db.map {
  total2 += r.value
}
assert(total2 == 19, "Sum after key removal: {total2}");
}
```

16. Hash

A 'hash' lets you find a record by its key in the same time whether you have 10 records or 10 million — there is no searching through a list, just a direct jump to the answer. Unlike 'sorted' and 'index', a hash has no meaningful order — you use it purely for fast lookups. List the key fields in brackets: 'hash<Type[field]>'. Combine a hash with a vector when you want both fast lookup and a stable iteration order.

```
struct Keyword {
  name: text
}
```

```
struct Data {
  h: hash < Keyword[name] >
}
```

16.0.1. Combining Hash and Vector

Pairing a hash with a vector on the same record type gives you the best of both worlds: the hash for instant lookup by key, and the vector for iterating in insertion order.

```
struct Count {
  t: text,
  v: integer
}
```

```
struct Counting {
  entries: vector < Count >,
  lookup: hash < Count[t] >
}
```

```
fn fill(c: Counting) {
  c.entries =[
 {t: "One", v: 1 },
 {t: "Two", v: 2 },
 {t: "Three", v: 3 },
 {t: "Four", v: 4 }
  ]
}
```

```
fn main() {
```

16.0.2. Basic Lookup

Fill a hash the same way you fill a vector. Look up by key with square brackets. A found record is truthy; a missing key returns null.

```
d = Data { };
d.h =[ {name: "one" }, {name: "two" }];
d.h +=[ {name: "three" }, {name: "four" }];
assert(d.h["three"], "Key exists");
assert(!d.h["None"], "Missing key returns null");
```

16.0.3. Hash + Vector Together

Here 'entries' is the vector (keeps insertion order) and 'lookup' is the hash (fast access). Both fields point to the same records — adding to one automatically updates the other.

```
c = Counting { };
fill(c);
assert(c.lookup["Three"].v == 3, "Hash lookup: Three");
assert(c.lookup["One"].v == 1, "Hash lookup: One");
assert(!c.lookup["Five"], "Missing key returns null");
```

Iterate in insertion order via the vector; look up by name via the hash. The vector also supports #first, #count, and #remove inside the loop.

```
add = 0;
for item in c.entries {
  add += item.v;
}
assert(add == 10, "Sum via vector iteration: {add}");
```

16.0.4. Removing a Key

Assigning null to a hash subscript removes that element. Removing a key that is not present is a safe no-op.

```
d.h["three"] = null;
assert(!d.h["three"], "three was removed");
assert(d.h["one"], "one still present");
d.h["missing"] = null;
```

no-op; does not panic

16.0.5. Why you cannot iterate a hash directly

A hash has no stable element order — internally, each record is placed in a slot chosen by a number computed from the key, so iteration order is unpredictable. 'for item in c.lookup' is therefore a compile error. Always pair a hash with a vector when you need both fast lookup and ordered iteration.

```
}
```

17. File

A file handle lets you read and write files without worrying about when the OS opens or closes them. The file opens on the first read or write and closes automatically when the handle goes out of scope.

17.0.1. Inspecting the File System

`file(path)` creates a File handle without opening anything yet. `f\#format` tells you what kind of path you are looking at: TextFile (plain text), LittleEndian (binary, bytes stored least-significant first — common on most PCs), BigEndian (binary, bytes stored most-significant first — common in network protocols), Directory, or NotExists. `lines()` reads a text file and returns it as a vector of lines. `exists(path)` is a convenience shorthand for checking that a path is not NotExists. `delete(path)` removes a file and returns a `FileResult` enum; use `.ok()` to check success. Clean up any leftover files from a previous interrupted run.

```
fn cleanup() {
  delete("test.bin");
  delete("test2.bin");
  delete("buffer.bin");
}
```

```
struct Buffer {
  size: i32,
  data: vector < single >
}
```

```
fn main() {
  cleanup();
```

Asking for the format of a directory path returns Directory, not a file format. `lines()` reads the named text file and splits it on newlines.

```
  ex = file("tests/example");
  assert(ex#format == Directory, "example is a directory");
  c = file("tests/example/config/terrain.txt").lines();
  assert(c[1] == "    terrain = [", "Line was '{c[1]}'");
```

`exists` and `delete` are safe to call when the file is absent. `delete` returns false rather than crashing when nothing is there.

```
  assert(!exists("test.bin"), "File should not exist before the test.");
  assert(!delete("nonexistent_xyz.bin").ok(), "delete on missing file not ok");
```

17.0.2. Writing a Text or Binary File

Wrapping the file handle in a block ensures the file closes the moment the block ends. Set `f\#format` to LittleEndian or BigEndian before writing binary data. Use `f += value` to append the raw bytes of any scalar value (u8, u16, i32, long, single, float, or text).

```
{f = file("test.bin");
 assert(f#format == NotExists, "File should not exist yet.");
 f#format = BigEndian;  // BigEndian: most-significant byte written first.
 f += 0 as u8;
 f += 1 as u8;
 f += 0x203 as u16;
 f += 0x4050607;
 f += 0x8090a0b0c0d0e0fl;
```

`f\#size` returns the total number of bytes written so far.

```
 assert(f#size == 16l, "Should have written 16 bytes.");
```

Text is written as raw UTF-8 bytes with no length prefix.

```
 f += "Hello world!";
 assert(f#size == 28l, "Size should be 28 bytes (16 + 12).");
}  // The file closes when the handle goes out of scope.
```

`move(src, dst)` renames a file. It refuses if the destination already exists or if either path would leave the project directory.

```
 assert(exists("test.bin"), "File should exist after writing.");
 assert(!move("test.bin", "../test.bin").ok(), "Should refuse to move outside
the project.");
 assert(move("test.bin", "test2.bin").ok(), "Could not move the file.");
```

17.0.3. Reading Back What You Wrote

When you open an existing file the default format is TextFile. Set `f\#format` to match the format used when writing before you read any bytes. `f\#read(n) as T` reads exactly n bytes and interprets them as type T. `f\#index` is the byte offset where the last read started. `f\#next` is the current read position; assign to it to seek anywhere.

```
{f = file("test2.bin");
 assert(f#format == TextFile, "The default format is TextFile.");
 f#format = LittleEndian;
```

The file was written as BigEndian, so reading the same 4 bytes as LittleEndian produces a byte-swapped value — that is intentional here and confirms that the bytes were stored in the order you chose.

```
 v = f#read(4) as i32;
 assert(v == 0x3020100, "BigEndian bytes 0..3 read as LittleEndian i32.");
 assert(f#index == 0l, "Last read started at byte 0.");
 assert(f#next == 4l, "Next read starts at byte 4.");
```

Seek to byte 16 to skip past the integers and read the text directly.

```
  f#next = 16l;
  s = f#read(5) as text;
  assert(s == "Hello", "Partial text read from offset 16.");
  assert(f#index == 16l, "Last read started at byte 16.");
  assert(f#next == 21l, "Next position after 5-byte text read.");
```

Requesting more bytes than remain simply returns whatever is left.

```
  rest = f#read(100) as text;
  assert(rest == " world!", "Read continues to end of file.");
  assert(f#next == f#size, "Position should be at end of file.");
```

You can seek back to any position and re-read.

```
  f#next = 0l;
  assert(f#read(4) as i32 == 0x3020100, "Seek-and-reread matches original.");
 }
  assert(delete("test2.bin").ok(), "Could not remove the test file.");
```

17.0.4. Working with Vectors and Struct Data

`f += vector\<T\>` writes every element in sequence as raw bytes, which is the fastest way to dump a whole collection to disk. Setting `f\#size = n` truncates or zero-extends the file to exactly n bytes — useful for discarding the tail after you have finished writing.

```
 {f = file("buffer.bin");
  f#format = LittleEndian;
  ints =[1, 2, 3, 4];
  f += ints;
  assert(f#size == 16l, "Four i32 values = 16 bytes");
```

Truncate to the first two integers.

```
  f#size = 8l;
  assert(f#size == 8l, "Truncated to 8 bytes");
 }
  assert(delete("buffer.bin").ok(), "Could not remove buffer.bin after vector
write test.");
```

Write a count followed by individual float values. `sizeof(T)` returns the byte width of a type, so you can compute the correct read length without hard-coding magic numbers.

```
 {buf = file("buffer.bin");
  buf#format = LittleEndian;
  buf += 4 as i32;
  buf += 1.1f;
  buf += 1.2f;
  buf += 2.1f;
```

```
 buf += 2.2f;
}
```

Read the count, then use it to read exactly that many floats directly into a struct field. This pattern lets you serialise and deserialise structs with variable-length data cleanly.

```
{b = Buffer { };
 f = file("buffer.bin");
 f#format = LittleEndian;
 n = f#read(4) as i32;
 b.data = f#read(n * sizeof(single));
}
 assert(delete("buffer.bin").ok(), "Could not remove buffer.bin.");
```

17.0.5. Error handling

File operations that can fail return a `FileResult` enum. Call `.ok()` to check success. The program does not crash on failure — you decide what to do.

```
 result = delete("this_file_does_not_exist.txt");
 assert(!result.ok(), "delete missing file returns not-ok");
```

Reading a non-existent file produces an empty result, not a crash.

```
 ghost = file("no_such_file.txt");
 assert(ghost#format == NotExists, "non-existent file has NotExists format");
```

`move` refuses paths outside the project directory.

```
 assert(!move("any.txt", "../escape.txt").ok(), "move outside project fails");
```

Writing to a read-only or invalid path also returns a FileResult. Always check `.ok()` after `delete`, `move`, `mkdir`, and `mkdir_all`.

```
}
```

18. Lexer

The lexer library breaks a text into tokens so your program can understand its structure. Tokens are the smallest meaningful pieces: numbers, names (like variable and function names), operators, and quoted strings. You tell the lexer which multi-character sequences count as one token and which words are reserved keywords, and it handles the rest.

18.0.1. Setting Up the Lexer

Create a `lexer::Lexer` and register your language's rules before you parse anything. `set_tokens` ensures operators like `+=` or `\>\>` are scanned as one token instead of two separate characters. `set_keywords` prevents reserved words from being treated as ordinary names — the lexer will report them exactly as written so your parser can treat them specially.

```
use lexer;
fn main() {
  l = lexer::Lexer { };
  l.set_tokens(["+=", "*=", "-=", "<=", ">=", "!=", "==", ">>", "<<", "->", "=>",
">>>", "..", "..=", "&&", "||"]);
  l.set_keywords(["for", "in", "if", "else", "fn", "pub", "use", "struct",
"enum", "match", "and", "or"]);
```

18.0.2. Reading Tokens

`parse_string(name, source)` feeds source text into the lexer. The name is used in error messages and position reports. After that, call the typed reader functions one by one to consume tokens in order.

`int()` consumes and returns the next integer token, or null if the current token is not an integer. `long_int()` does the same for integers suffixed with `l`. `matches(s)` consumes the next token only when it equals s and returns true; otherwise it leaves the token in place and returns false. `peek()` returns the next token as text without consuming it. `position()` returns the current location as `file:line:col`.

```
  l.parse_string("Tokens", "12 += -2 * 3l >> 4");
  assert(l.int() == 12, "Integer");
  assert(!l.matches("+"), "Incorrect plus");
  assert(l.peek() != "+", "Incorrect plus");
  assert(l.matches("+="), "Incorrect plus_is");
  assert(l.int() == -2, "Second integer");
  assert(l.matches("*"), "Incorrect multiply");
  assert(l.int() != 3, "Third number");
  assert(l.long_int() == 3, "Incorrect long");
  assert(l.position() == "Tokens:1:15", "Incorrect position {l.position()}");
  assert(!l.matches(">"), "Incorrect higher");
  assert(l.matches(">>"), "Incorrect logical shift");
  assert(l.position() == "Tokens:1:18", "Incorrect position {l.position()}");
```

18.0.3. String Literals and Comments

`constant_text()` reads a double-quoted string and handles special codes like \n (newline) and \\ (backslash). `constant_character()` reads a single-quoted character literal and returns it as text.

```
  l.parse_string("Texts", "\"123\" + '4'");
  assert(l.constant_text() == "123", "Incorrect text literal");
  assert(l.matches("+"), "Incorrect add");
```

P111: constant_character() returns a character, not text “123”. This assert was passing only because char == text always returned true. TODO: fix the test to match actual constant_character() semantics. assert(“{l.constant_character()}” == “123”, “Incorrect text literal”); The lexer collects `//` comments automatically as it scans. You do not need to handle them yourself. `last_comment()` returns the accumulated comment text since the last consumed token. When multiple comment lines appear in a row they are joined with newlines into a single string. `comment_behind()` is true when the comment appeared on the same line as the preceding token rather than on its own line above. `is_finished()` returns true once every token has been consumed.

```
  l.parse_string("Comments", "// starting comments\n123 // same line comment\n//
extra comment\n4");
  assert(!l.comment_behind(), "Initial comment not behind");
  assert(l.last_comment() == "starting comments", "Initial comment");
  assert(l.int() == 123, "Content integer");
  assert(l.comment_behind(), "Second comment is behind");
  assert(l.last_comment() == "same line comment\nextra comment", "Second
comment");
  assert(!l.is_finished(), "Not Ready");
  assert(l.int() == 4, "Second integer");
  assert(l.last_comment() == "", "No remaining comment");
  assert(l.is_finished(), "Ready");
```

18.0.4. Embedded Format Expressions

Loft string literals can embed expressions with `{expr}`. The lexer exposes a protocol that lets you parse these yourself. When `constant_text()` reaches a `{`, it returns the literal text before it and sets `is_formatting()` to true. At that point call `set_formatting(false)` and parse the embedded expression normally using the usual token readers. When the expression is done, call `set_formatting(true)` and consume the closing `}}`. Then `constant_text()` continues with the next segment of the string.

```
  l.parse_string("Formatting", "\"abc{{12 + 34}}def\"");
  assert(l.constant_text() == "abc", "Before formatting");
  assert(l.is_formatting(), "Formatting");
  l.set_formatting(false);
  assert(l.int() == 12, "First integer");
  assert(l.matches("+"), "Incorrect plus");
  assert(l.int() == 34, "Second integer");
  l.set_formatting(true);
  assert(l.matches("}}"), "Incorrect closing brace");
  assert(l.constant_text() == "def", "After formatting");
  assert(!l.is_formatting(), "Formatting");
}
```

19. Parser

The ‘parser’ library lets a Loft program read and validate other Loft source code at runtime. This is useful when you want to:

- validate configuration files written in the Loft syntax
- build tools that inspect or transform Loft source
- write a test that checks whether a generated code snippet is syntactically correct

Together the ‘lexer’ and ‘parser’ libraries are designed as a reusable foundation. You can use them to build entirely new languages: feed the lexer your own token rules, then layer a custom parser on top. This keeps the tokeniser and grammar logic separate from the language runtime so you only bring in what you need.

‘parse(name, source)’ is the single entry point. It takes a display name (used in error messages) and a Loft source string. The name does not need to correspond to a real file — it is only used when reporting parse errors.

If the source is invalid, parse() emits a diagnostic error and the call returns without producing a value.

19.0.1. What the parser understands

The parser handles all Loft syntax:

- ‘struct Name { field: type [= default] }’ — data containers with named fields
- ‘enum Name { Variant [{ field: type }] }’ — named choices, each with optional data
- ‘fn name(params) [-> type] { body }’ — functions with a block body
- ‘fn name(params) [-> type]; #rust “template”’ — operator templates backed by Rust
- ‘use module;’ — module imports
- Expressions: binary operators with precedence, function calls, field access,

```
index expressions, if/else, for loops, blocks, and formatted string literals
```

- Type expressions: plain names, generic types like ‘vector<T>’, keyed

```
collections (sorted/hash/index), and integer ranges with 'limit(min, max)'
```

```
use parser;
fn main() {
```

19.0.2. Parsing a minimal function

The double braces ‘{{‘ and ‘}}’ produce literal ‘{‘ and ‘}’ inside a Loft format string — needed here because the source code itself contains braces.

```
  parser::parse("hello", "fn main() {{ println(\"Hello World\"); }}");
```

19.0.3. Parsing a struct and function together

The parser validates the entire source snippet as one unit. Both the struct definition and the function body are checked.

```
  parser::parse("point", "struct Point {{ x: integer, y: integer }} fn origin() -
> Point {{ Point {{ x: 0, y: 0 }} }}");
```

19.0.4. Parsing an enum

Both a plain variant and a struct variant (with fields) are recognised.

```
  parser::parse("shape", "enum Shape {{ Circle {{ radius: float }}, Rectangle
{{ w: float, h: float }} }}");
```

19.0.5. Parsing a for loop and arithmetic

This exercises the expression parser and range syntax.

```
  parser::parse("loop", "fn sum(n: integer) -> integer {{ t = 0; for i in 1..=n
{{ t += i; }} t }}");
```

19.0.6. Practical use: validating user-supplied code

If your application lets users write Loft snippets (for scripting or configuration), you can parse them before executing:

```
snippet = read_user_input();
parser::parse("user_code", snippet);
// If parse() emits errors, the snippet was invalid — show them to the user.
```

Combine this with the lexer (see the Lexer page) when you need to extract individual tokens from the source rather than validate all of the syntax.

```
  println("parser test passed");
}
```

20. Libraries

A library is a `.loft` file you can share across projects. Place it in a `lib/` directory, import it with `use name;` at the top of your file, and then refer to its types and functions using the `name::` prefix. All types and functions in a library are accessible — there is no way to mark something as internal or hidden. `use` statements must come before any `fn` or `struct` definition in your file. Putting a `use` after a definition is a syntax error.

```
use testlib;
```

20.0.1. Constants and Enums

Library constants and enum variants are accessed with the `libname::` prefix. You can compare, order, and convert enum values exactly as you would with enums defined in the same file.

20.0.2. Struct Construction and Field Access

Construct a library struct with its full namespace prefix. Omitting the prefix is a parse error. Once you have a value, field access uses the plain dot notation with no prefix needed.

20.0.3. Calling Library Methods

Methods defined in the library are called on the value directly — no prefix on the call site. Free functions (not methods) do need the `libname::` prefix.

20.0.4. Extending a Library Type

You can add new methods to a library type from your own file. Write the `self` parameter type with the `::` separator and the compiler treats the function as a method on that type.

```
fn shifted(self: testlib::Point, dx: float, dy: float) -> testlib::Point {
  testlib::Point {x: self.x + dx, y: self.y + dy }
}
```

```
fn main() {
```

Constants from a library require the library prefix.

```
  assert(testlib::MAX_SIZE == 100, "library constant MAX_SIZE");
  assert(testlib::MIN_SIZE == 1, "library constant MIN_SIZE");
```

Library enum variants: comparison and ordering work as normal.

```
  s = testlib::Ok;
  assert(s == testlib::Ok, "enum equality");
  assert(s < testlib::Error, "enum ordering");
  assert("{s}" == "Ok", "enum to text: {s}");
  assert("Warning" as testlib::Status == testlib::Warning, "text to enum");
```

Library enum values can be passed to library free functions.

```
assert(testlib::status_text(testlib::Error) == "err", "free fn with enum arg");
```

Structs are constructed with namespace prefix and accessed by field.

```
p = testlib::Point {x: 3.0, y: 4.0 };
assert(p.x == 3.0, "field access: x={p.x}");
assert(p.y == 4.0, "field access: y={p.y}");
```

Methods defined in the library are called without a prefix.

```
dist = p.distance();
assert(round(dist) == 5.0, "method call: distance={dist}");
```

User-defined method on a library type (self: testlib::Point) works.

```
assert(p.shifted(1.0, 2.0).x == 4.0, "shifted x");
assert(p.shifted(0.0, 2.0).y == 6.0, "shifted y");
```

Scalar fields can be mutated directly.

```
b = testlib::Bag {label: "test", count: 0 };
b.label = "updated";
assert(b.label == "updated", "direct scalar field mutation");
```

Methods that mutate scalar fields work.

```
b.bump();
assert(b.count == 1, "method scalar mutation: count={b.count}");
b.bump();
assert(b.count == 2, "method scalar mutation: count={b.count}");
```

Free functions: called with the library prefix.

```
assert(testlib::add(3, 4) == 7, "free function add");
assert(testlib::add(0, -5) == -5, "free function negative");
```

Library types work as function parameter types when written with their full namespace prefix in the function signature. A struct with a vector field can be initialised via a struct literal, including elements that are themselves library structs.

```
bag2 = testlib::Bag {label: "full", count: 3, items: [
testlib::Point {x: 1.0, y: 0.0 },
testlib::Point {x: 0.0, y: 1.0 }
] };
assert(len(bag2.items) == 2, "initialized vector field: {len(bag2.items)}");
assert(bag2.items[0].x == 1.0, "vector field element access");
```

Appending a struct variable with `+= \[var\]` and a whole vector with `+= vec` both work.

```
  extra = testlib::Point {x: 7.0, y: 8.0 };
  bag2.items +=[extra];
  assert(len(bag2.items) == 3, "append via var: len={len(bag2.items)}");
  assert(bag2.items[2].x == 7.0, "append via var: x={bag2.items[2].x}");
  more_pts =[testlib::Point {x: 9.0, y: 10.0 }, testlib::Point {x: 11.0, y:
12.0 }];
  bag2.items += more_pts;
  assert(len(bag2.items) == 5, "append vector: len={len(bag2.items)}");
  assert(bag2.items[3].x == 9.0, "append vector[3].x={bag2.items[3].x}");
```

Importing the same library twice is silently ignored. A library can itself import other libraries using `use`, so dependency chains work.

20.0.5. Package Layout and loft.toml

A library can be distributed as a directory package instead of a single flat file. The packaged directory layout is:

```
mylib/
  loft.toml           optional manifest
  src/
    mylib.loft        library source (default entry)
```

When the interpreter searches a lib directory and finds `\<dir\>/mylib/` it looks for `\<dir\>/mylib/src/mylib.loft` automatically.

The optional `loft.toml` manifest supports two settings:

```
[package]
loft = ">=1.0"        minimum interpreter version required
```

```
[library]
entry = "src/mylib.loft"   override the default entry path
```

If the interpreter version is below the stated minimum, loading the library produces a fatal compile error describing the version mismatch. If no manifest is present, the default entry `src/\<name\>.loft` is used.

20.0.6. Wildcard and Selective Imports

By default, library names require the `libname::` prefix. You can import names directly into your namespace with `use lib::\*` (wildcard) or `use lib::Name, Other` (selective). Only `pub`-marked definitions in the library are imported this way. Non-pub definitions remain accessible via the `lib::name` prefix.

20.0.7. Limitations

`use` must appear before all definitions. If you write a function first and then a `use`, the compiler reports a syntax error.

`pub` controls import visibility. Only `pub`-marked definitions are available via wildcard (`use lib::\*`) or selective import. Non-pub definitions are still accessible with the `lib::name` prefix.

No native extension loading. Libraries are pure `.loft` files. Native Rust extensions (`loft.toml` with `native = "..."`) are planned for a future release.

```
}
```

21. Store Locks

Loft gives you two separate ways to protect a value from accidental changes:

1. **Compile-time const**: mark a variable or parameter with 'const' and the

```
compiler refuses to compile any code that tries to reassign it. The check
happens before the program even runs — zero runtime cost.
```

2. **Runtime lock**: the '#lock' attribute on a reference lets you lock a

```
store at runtime so that any write attempt panics immediately, even across
function boundaries. This is useful for debugging: turn it on when you
suspect an unexpected mutation.
```

```
struct Counter {
  value: integer
}
```

21.0.1. const parameters

'const' on a parameter is a compile-time promise: "this function will not modify this value." The compiler enforces it — any assignment to a const parameter is a compile error, caught before you run anything.

```
fn read_value(self: const Counter) -> integer {
  self.value
}
```

A non-const parameter leaves the store unlocked so the function can write.

```
fn increment(self: Counter) {
  self.value += 1
}
```

```
fn main() {
```

21.0.2. const local variables

Declare a local variable with 'const' to signal that it will not change after its first assignment. The compiler rejects any later assignment to it — reassigning or appending to a const variable is a compile error.

This is handy for configuration values or lookup tables that should never be overwritten by accident deep inside a long function.

```
  const limit = 100;
  assert(limit == 100, "const integer is readable");
```

const also works for struct references.

```
const cfg = Counter {value: 42 };
assert(cfg.value == 42, "const reference is readable");
```

Passing a const reference to a const parameter is always allowed.

```
assert(read_value(cfg) == 42, "const passed to const param");
```

21.0.3. Calling methods on const references

A non-const method can still be called on a non-const variable even after you have manually locked the store; the lock is a runtime check.

```
c = Counter {value: 10 };
increment(c);
assert(c.value == 11, "increment modified c");
```

21.0.4. Runtime store locks with #lock

'#lock' is an attribute on any reference variable. Setting it to true turns on a runtime guard: any write to that store will panic immediately, wherever it happens. A freshly created reference starts unlocked.

```
d = Counter {value: 5 };
assert(!d#lock, "new store starts unlocked");
d#lock = true;
assert(d#lock, "store is locked after assignment");
```

You can still read from a locked store — only writes are blocked.

```
assert(read_value(d) == 5, "locked store is still readable");
```

21.0.5. When to use each approach

- Use 'const' on parameters and locals to express your design intent

```
and get compile-time safety at zero cost.
```

- Use '#lock = true' when you want a runtime tripwire: you suspect some

```
code path is mutating a value it should not touch, and you want the
program to panic with a precise location rather than corrupt silently.
```

get_store_lock() is the function form of the #lock attribute. Both return the same boolean.

```
  e = Counter {value: 99 };
  assert(get_store_lock(e) == e#lock, "function form matches attribute before
lock");
  e#lock = true;
  assert(get_store_lock(e) == e#lock, "function form matches attribute after
```

```
lock");
}
```

22. Parallel execution

The `par(b=worker_call, threads)` clause on a `for` loop runs a function on every element of a vector in parallel and gives you the results one by one in the loop body.

22.0.1. Why parallel loops?

When you have a large collection and each element can be processed independently — image filters, score calculations, data transforms — `par()` splits the work across CPU cores automatically. You write a normal for loop; adding `par(...)` makes it parallel with no other changes.

22.0.2. Syntax

```
`for a in vec par(b=func(a), N) { body }`
```

- `a` — loop variable, read-only reference to the current element
- `b` — result variable, holds the return value of `func(a)`
- `func(a)` — worker function called on each element
- `N` — number of threads (1 = sequential, 4 = typical)

22.0.3. Two call forms

- **Form 1** — global function: `par(b=my_func(a), 4)`
- **Form 2** — method on element: `par(b=a.my_method(), 4)`

22.0.4. Worker function rules

The worker function:

- Takes a `const` reference to the element (read-only, no mutation)
- Returns a value (integer, float, boolean, text, or a struct)
- Must not use global state or I/O (no println, no file access)
- Can accept extra arguments forwarded from the calling scope

Results are delivered in the original order regardless of which thread finishes first.

```
struct Score {
  value: integer
}
```

```
struct ScoreList {
  items: vector < Score >
}
```

```
struct DoubledScore {
  label: text,
  doubled: integer
}
```

```
fn double_score(thr_r: const Score) -> integer {
  thr_r.value * 2
}
```

```
fn scale_score(thr_r: const Score, thr_factor: integer) -> integer {
  thr_r.value * thr_factor
}
```

```
fn make_doubled(thr_r: const Score) -> DoubledScore {
  DoubledScore { label: "v{thr_r.value}", doubled: thr_r.value * 2 }
}
```

```
fn get_value(self: const Score) -> integer {
  self.value
}
```

```
fn make_scores() -> ScoreList {
  thr_q = ScoreList { };
  thr_q.items +=[Score {value: 10 }, Score {value: 20 }, Score {value: 30 }];
  thr_q
}
```

```
fn main() {
```

22.0.5. Global Function (Form 1)

Each Score's value is doubled by `double_score` across 4 threads. The loop body sees `b` with the doubled value, in original order.

```
  q = make_scores();
  sum = 0;
  for thr_a in q.items par(thr_b = double_score(thr_a), 4) {
    sum += thr_b;
  }
```

10*2 + 20*2 + 30*2 = 120

```
  assert(sum == 120, "parallel double: sum == 120");
```

22.0.6. Extra Arguments

The worker function can take extra arguments from the calling scope. Here `scale_score(a, factor)` passes `factor=3` to every worker.

```
  q1b = make_scores();
  factor = 3;
  scaled_sum = 0;
```

```
for thr_a in q1b.items par(thr_b = scale_score(thr_a, factor), 4) {
  scaled_sum += thr_b;
}
```

10*3 + 20*3 + 30*3 = 180

```
assert(scaled_sum == 180, "extra arg: scaled sum == 180");
```

22.0.7. Struct Return

Workers can return a struct. Text fields are deep-copied so they remain valid after the worker thread exits.

```
q2 = make_scores();
labels = "";
for thr_sa in q2.items par(thr_ds = make_doubled(thr_sa), 1) {
  labels += "{thr_ds.label},";
}
assert(labels == "v10,v20,v30,", "struct return: {labels}");
```

22.0.8. Method Call (Form 2)

`b=a.get_value()` dispatches the method on each element in parallel. This is syntactic sugar — equivalent to `b=get_value(a)`.

```
q3 = make_scores();
total = 0;
for thr_a in q3.items par(thr_b = thr_a.get_value(), 4) {
  total += thr_b;
}
assert(total == 60, "method call: total == 60");
```

22.0.9. Empty Vector

An empty vector is safe — the loop body never executes, no threads are spawned.

```
empty = ScoreList { };
thr_n = 0;
for thr_a in empty.items par(thr_b = double_score(thr_a), 1) {
  thr_n += 1;
}
assert(thr_n == 0, "empty: 0 iterations");
```

22.0.10. Sequential fallback

Using `par(..., 1)` runs the worker on a single thread — useful for debugging. The behaviour is identical; only the parallelism changes.

```
q4 = make_scores();
seq_sum = 0;
```

```
  for thr_a in q4.items par(thr_b = double_score(thr_a), 1) {
    seq_sum += thr_b;
  }
  assert(seq_sum == 120, "sequential par(1): same result");
}
```

23. Logging

Printing everything to the console is fine during development, but in a real application you usually want more control: write to a file, filter by severity, and keep the output even after the terminal session ends. Loft's logging functions give you that without changing your source code.

23.0.1. The four severity levels

Choose the level that matches how serious the event is:

- 'log_info' — routine progress; fine to see during development but often

```
silenced in production to keep log files small.
Example: "processing file X", "connected to database".
```

- 'log_warn' — something unexpected happened but the program recovered.

```
Example: "config key missing, using default", "retrying after timeout".
```

- 'log_error' — something went wrong and you need to investigate, but the

```
program can continue (perhaps degraded).
Example: "failed to save record", "unexpected null value".
```

- 'log_fatal' — a condition so serious that normal operation is impossible.

```
Example: "cannot open database", "required config file not found".
```

23.0.2. Configuring the log destination

By default, log calls do nothing — no file is written, no console output is produced. To switch logging on, place a 'log.conf' file in the same directory as your '.loft' file, or pass '–log-conf path/to/log.conf' on the command line.

Generate a documented template with all defaults by running:

```
loft --generate-log-config
```

A minimal 'log.conf' looks like this:

```
[log]
file  = log.txt    # write messages here (relative to the .loft file)
level = info       # minimum level to record; choices: info warn error fatal
```

```
[rotation]
max_size_mb = 500  # start a new log file after this many megabytes
daily       = true # also start a new file at midnight UTC
max_files   = 10   # keep at most this many log files (older ones are deleted)
```

```
[rate_limit]
per_site = 5       # suppress messages from the same source line after 5/minute
```

```
[levels]
# Override the global level for a specific file:
# "debug_tool.loft" = info
# "src/"            = error
```

23.0.3. Production mode

When ‘production = true’ is set in log.conf:

- ‘panic()’ becomes a fatal log entry instead of aborting the process.
- A failing ‘assert()’ becomes an error log entry instead of aborting.

The program keeps running and the problem is captured in the log — useful for long-running services where a single error should not bring everything down.

23.0.4. Using format strings in log messages

Log messages are plain text, but you can embed any expression using the same ‘{...}’ format syntax as everywhere else in Loft. The interpolation happens only when the message is actually going to be written; if the configured level is higher than the call, the string is never evaluated (no performance cost for suppressed messages).

```
fn main() {
```

Without a log.conf these calls do nothing — the tests below pass even though no output is produced.

```
  log_info("starting up");
  log_warn("this is a warning");
  log_error("something went wrong");
  log_fatal("critical failure");
```

23.0.5. Example log.txt output

When a ‘log.conf’ is present with ‘level = info’, the four calls above produce entries like this in ‘log.txt’:

```
2026-03-24 09:15:00 INFO   app.loft:3  starting up
2026-03-24 09:15:00 WARN   app.loft:4  this is a warning
2026-03-24 09:15:00 ERROR  app.loft:5  something went wrong
2026-03-24 09:15:00 FATAL  app.loft:6  critical failure
```

Each line contains: timestamp, severity level, source file and line number, then the message. This makes it easy to search the file for errors or trace back to the exact line that produced a message. A true assert never logs anything — it is only the false case that logs.

```
  assert(true, "this should never fail");
```

23.0.6. Typical usage pattern

In a real program you would write something like:

```
fn process(item: Item) {
    log_info("processing item {item.id}");
    result = do_work(item);
    if !result {
        log_error("work failed for item {item.id}");
        return;
    }
    log_info("finished item {item.id} successfully");
}
```

The log messages give you a record of exactly what the program did and where it went wrong, without cluttering the terminal during normal runs.

```
  println("logging test passed");
}
```

24. Time

Loft provides two time functions:

```
now()     — milliseconds since the Unix epoch (wall-clock time).
ticks()   — microseconds elapsed since program start (monotonic clock).
```

Both return a 'long'.

Use 'now()' for timestamps, log entries, and date calculations. Use 'ticks()' for benchmarks and frame timing — it is unaffected by system clock changes or NTP adjustments.

```
fn main() {
```

24.0.1. Wall-clock time: now()

'now()' returns the current time as milliseconds since 1970-01-01T00:00:00 UTC. The value is always positive and grows over time.

```
  t = now();
  assert(t > 0l, "now() must be positive");
```

Two successive calls return non-decreasing values.

```
  t2 = now();
  assert(t2 >= t, "now() must be non-decreasing");
```

24.0.2. Elapsed time: ticks()

'ticks()' measures microseconds since the program started. It uses a monotonic clock so it never jumps backward.

```
  start = ticks();
  assert(start >= 0l, "ticks() must be non-negative");
  end = ticks();
  assert(end >= start, "ticks() must be monotonically non-decreasing");
```

24.0.3. Measuring elapsed time

Subtract two 'ticks()' values to get the duration in microseconds. Divide by 1000 to convert to milliseconds.

```
elapsed_us  = end_ticks - start_ticks
elapsed_ms  = elapsed_us / 1000l
```

Example: measure how long a loop takes.

```
  t_before = ticks();
  sum = 0;
  for i in 0..1000 {
```

```
    sum += i
  }
  t_after = ticks();
  elapsed = t_after - t_before;
  assert(elapsed >= 0l, "elapsed time must be non-negative: {elapsed}");
  assert(sum == 499500, "loop produced wrong sum: {sum}");
```

24.0.4. Common patterns

Timestamp a log entry (seconds since epoch):

```
seconds = now() / 1000l
```

Seed the random number generator with the current time:

```
rand_seed(now() as integer)
```

Simple stopwatch:

```
start = ticks()
... do work ...
log_info("Done in {(ticks() - start) / 1000l} ms")
```

```
}
```

25. Safety

Loft catches many errors at compile time, but a few surprises remain at runtime. This page catalogues every known trap so you can write confident code from day one. Each section includes a live example that proves the described behavior. Helper used in the '??' double-evaluation example below. Increments the call counter and returns the given value unchanged.

```
fn counted_call(calls: &integer, value: integer) -> integer {
  calls += 1;
  value
}
```

Used to obtain a null text value (an uninitialized field is the NUL sentinel).

```
struct NullTextHolder { s: text }
```

```
fn main() {
```

25.0.1. Null values — hidden reserved values

Every type reserves one special value to mean "nothing here" (null). That reserved value looks like any other value, so be aware of what it is for each type:

- `boolean` — `false` is the null value
- `integer` — the most negative 32-bit integer (-2 147 483 648) is null
- `long` — the most negative 64-bit integer is null
- `float/single` — `NaN` (Not a Number) is null
- `character` — the NUL character (code point 0) is null
- `text` — a single NUL character (`'\\0'`) is null; the empty string `""` is NOT null
- `reference` — record 0 is null
- plain `enum` — byte value 255 is null (limits enums to 255 variants)

This means there is one value per type that you cannot distinguish from null. For integers, that is −2 147 483 648. Division by zero also produces this same value, so both paths look the same to your code:

```
  zero = 0;
  n = 1 / zero;
  assert(!n, "div-by-zero is null");
  assert(n != 0, "null is not zero; it is -2 147 483 648");
  assert(n < 0, "i32::MIN is the most negative 32-bit integer");
```

Arithmetic on null propagates: null plus anything is null.

```
  assert(!(n + 1), "null + 1 is still null");
```

Text null is the NUL character ('\0'), not the empty string. Parsing a JSON object with a missing text field produces the NUL sentinel:

```
holder = NullTextHolder.parse(`{{}}`);
assert(!holder.s, "missing JSON text field is null (the NUL sentinel)");
empty = "";
assert(empty, "empty string is NOT null — this surprises most newcomers");
```

Mitigation: Use `long` when you need the full 32-bit integer range, or declare struct fields as `not null` so that reserved value can be used as data.

25.0.2. Integer overflow wraps silently

32-bit integers wrap when they exceed roughly 2 billion. There is no runtime overflow check and no exception. The result is a small or negative number and the program continues as if nothing happened. The following would wrap silently in a release build:

```
big = 2000000000; big + big  →  negative number
```

Mitigation: Use `long` (64-bit) when multiplying or summing large values: `big as long + big as long` avoids the wrap. Design: a future `--debug` flag will turn silent overflow into a logged WARN with file:line context so overflows are visible during development without affecting release performance.

```
big = 2000000000;
assert(big as long + big as long == 4000000000l, "long avoids wrap");
```

25.0.3. Bitwise operators with zero

All bitwise operators (AND, OR, XOR, shift) work correctly with zero. Zero is the identity element for OR, XOR, and shift; zero for AND.

```
assert(0b1010 & 0 == 0, "AND with 0: zero");
assert(0b1010 | 0 == 0b1010, "OR with 0: identity");
assert(0b1010 ^ 0 == 0b1010, "XOR with 0: identity");
assert(5 << 0 == 5, "shift left by 0: identity");
assert(5 >> 0 == 5, "shift right by 0: identity");
```

25.0.4. Float null is NaN

Floats represent null as `NaN` (Not a Number). Null floats behave consistently with other null types: null is not equal to anything (including itself), and null is not-equal to everything.

```
bad = 0.0 / 0.0;
assert(!bad, "NaN is falsy — this is how you detect null floats");
assert(!(bad == bad), "null == null is false");
assert(bad != bad, "null != null is true");
assert(bad != 0.0, "null != 0.0 is true");
```

Use `!f` or `f ?? default` to check for null floats.

25.0.5. Text length counts bytes, not characters

`len()` on text returns the number of UTF-8 bytes, not the number of visible characters. Multi-byte characters (accented letters, emoji, CJK) each occupy 2-4 bytes. `size()` returns the number of Unicode code points (characters). These two values differ whenever the text contains any multi-byte character.

```
emoji = "Hi 😊!";
assert(len(emoji) == 8, "5 visible chars but 8 bytes (emoji is 4 bytes)");
assert(size(emoji) == 5, "size() counts code points: {size(emoji)}");
```

Slicing and indexing also use byte offsets. Slicing in the middle of a multi-byte character is an error. Mitigation: Use `for c in text` to iterate by character. Use `c\#index` and `c\#next` to get the byte boundaries of each character.

```
count = 0;
for c in emoji { count += 1; }
assert(count == 5, "for-loop iterates by character, not byte");
```

25.0.6. `\#index` means different things on text and vectors

On a text loop, `c\#index` is the byte offset of the current character. On a vector loop, `v\#index` is the 0-based element position. Both are called `\#index` but represent different units.

```
v = [10, 20, 30];
idx = 0;
for x in v { idx = x#index; }
assert(idx == 2, "vector #index: element position (0-based)");
```

Text #index is a byte offset, not a character count:

```
t = "aé";
byte_pos = 0;
for c in t { byte_pos = c#index; }
assert(byte_pos == 1, "text #index: byte offset of 'é' (byte 1, not char 1)");
```

25.0.7. `??` evaluates the left side exactly once

The `??` operator means "use this value, or if it is null, use the right side instead". The left-hand expression is evaluated exactly once regardless of whether the result is null. The example below uses `counted_call` (defined above) to verify this.

```
calls = 0;
qq_result = counted_call(calls, 7) ?? 99;
assert(calls == 1, "?? evaluated left side exactly once: {calls}");
assert(qq_result == 7, "result is the value from that single evaluation:
{qq_result}");
```

This means `result = expensive_call() ?? default` is safe: the function is called once. If it returns null, `default` is used. There is no double call.

25.0.8. Text indexing and slicing return different types

`txt\[i\]` returns a `character` (a single Unicode scalar value). `txt\[i..j\]` returns `text` (a UTF-8 string). These are different types.

```
txt = "hello";
ch = txt[0];
slice = txt[0..1];
assert(ch == 'h', "indexing returns a character");
assert(slice == "h", "slicing returns text");
```

Building text from characters requires format interpolation:

```
result = "";
for c in "abc" { result += "{c}"; }
assert(result == "abc", "characters must be formatted into text");
```

25.0.9. Format strings: braces are always interpreted

Every string literal in loft is a format string. Literal braces must be escaped as `{{` and `}}`:

```
n = 42;
assert("{n}" == "42", "single braces: format expression");
assert(len("{{}}") == 2, "double braces produce literal brace chars");
```

Forgetting to escape braces in expected output is a common mistake in assertions and comparisons.

25.0.10. Hash collections cannot be iterated

Hashes are lookup structures, not ordered collections. You cannot write `for item in my_hash { }`. If you need both fast lookup and ordered iteration, keep a vector and a hash pointing at the same record type. See the Hash documentation page for the recommended pattern.

25.0.11. Mutation guard blocks appending during iteration

The compiler prevents `v += \[x\]` inside `for e in v`. This protects against infinite loops. The guard also catches field access: `for e in db.items { db.items += \[x\]; }` is blocked too. The only allowed mutation is `e\#remove` inside a filtered loop.

25.0.12. If-expression requires else when used as a value

Using `if` as a value expression without an `else` clause is a compile error. This prevents accidental null values from missing branches. If-statements (where the body has no value) do not need else. For example, `x = if cond { 1 }` is an error; write `x = if cond { 1 } else { 0 }`.

25.0.13. Match guards do not prove a variant is handled

A guarded arm like `Red if cond =\> ...` does not count as handling the `Red` variant because the guard can fail at runtime. Even if every variant has a guarded arm, you still need a wildcard _ or an unguarded arm so the compiler knows every case is covered.

25.0.14. Ref-parameter semantics

Without &, appending to a vector parameter is local — the caller's vector does not grow. With &, the caller sees the new elements. Field-level mutations (e.g. `v\[i\].field = x`) are always visible

to the caller because both sides share the same underlying database reference. Rule of thumb: Use `&vector\<T\>` when the function needs to grow the vector. Use plain `vector\<T\>` when the function only reads or modifies existing elements.

25.0.15. File I/O assumes UTF-8

All file reading in loft assumes the file content is valid UTF-8. Reading a binary file or a file in a different encoding (e.g. Latin-1) will crash the program at runtime. There is currently no way to read raw bytes. Mitigation: Only read files you know to be UTF-8. If you need to process binary data, convert it to UTF-8 externally before passing it to loft.

25.0.16. XOR is ^, not exponentiation

Unlike some languages where ^ means "power", in loft ^ is bitwise XOR. The `\*\*` operator (Python/Ruby exponentiation) does not exist either. Use the `pow()` function for exponentiation.

```
  assert((0b1010 ^ 0b1100) == 0b0110, "^ is XOR");
  assert(pow(2.0, 3.0) == 8.0, "use pow() for power");
}
```

26. JSON

Loft has built-in JSON support: any struct can be serialised to JSON with a format flag, and parsed back from JSON text. No annotations or code generation needed — it works on every struct automatically.

26.0.1. Serialisation — struct to JSON

Use the ':j' format flag inside a format string to produce JSON output. Field names become quoted keys; strings are escaped; numbers and booleans are written as JSON literals.

```
"{my_struct:j}"
```

```
struct User {
  id: integer,
  name: text,
  email: text
}
```

26.0.2. Parsing — JSON to struct

Call 'Type.parse(text)' to create a struct from JSON text. Both JSON-style quoted field names ("name": value) and loft-style unquoted field names (name: value) are accepted. Missing fields get null sentinels. Extra fields in the input are silently skipped.

```
user = User.parse(json_text)
```

26.0.3. Vectors

Parse a JSON array into a vector of structs with 'vector<T>.parse(text)'.

```
scores = vector<Score>.parse("[{\"v\":1},{\"v\":2}]")
```

```
struct Score {
  value: integer
}
```

26.0.4. Parse Errors

After parsing, check '#errors' on the result to see what went wrong. Each error is a text string describing the problem and its location.

```
data = MyType.parse(bad_json)
for e in data#errors { log_warn(e) }
```

26.0.5. Nested Structs

Structs with struct-typed fields parse nested JSON objects automatically.

```
struct Address {
  city: text,
```

```
  zip: text
}
```

```
struct Contact {
  name: text,
  address: Address
}
```

```
fn main() {
  u = User { id: 42, name: "Alice", email: "alice@example.com" };
  json = "{u:j}";
  expected = `{{"id":42,"name":"Alice","email":"alice@example.com"}}`;
  assert(json == expected, "to json");
```

```
  bob = User.parse(`{{"id":7,"name":"Bob","email":"bob@test.org"}}`);
  assert(bob.id == 7, "parsed id: {bob.id}");
  assert(bob.name == "Bob", "parsed name");
  assert(bob.email == "bob@test.org", "parsed email");
```

```
  u2 = User.parse("{u:j}");
  assert(u2.id == u.id, "round-trip id");
  assert(u2.name == u.name, "round-trip name");
```

```
  bad = User.parse(`{{"id":"not_a_number","name":42}}`);
  errs = 0;
  for err in bad#errors {
    log_info("parse error: {err}");
    errs += 1;
  }
  assert(errs > 0, "expected parse errors, got {errs}");
```

```
  scores = vector<Score>.parse(`[{{"value":10}},{{"value":20}},{{"value":30}}]`);
  total = 0;
  for s in scores {
    total += s.value;
  }
  assert(total == 60, "vector sum: {total}");
```

```
  c = Contact.parse(`{{"name":"Carol","address":
{{"city":"Amsterdam","zip":"1012"}}}}`);
  assert(c.name == "Carol", "nested name");
  assert(c.address.city == "Amsterdam", "nested city: {c.address.city}");
  assert(c.address.zip == "1012", "nested zip");
```

26.0.6. Missing fields get null

When a field is absent from the JSON, the struct field gets its null sentinel. For text, that is the NUL character '\0' (not the empty string ""); for integer it is the minimum i32 value. Check with '!' before use, or assign a default with '??'.

```
  partial = User.parse(`{{"id":1}}`);
  assert(partial.id == 1, "partial id");
  assert(!partial.name, "missing name is null");
}
```

27. Generics

A generic function uses a type variable to work with any type. Write the function once; the compiler creates a specialised copy for each concrete type you call it with.

27.0.1. Declaring a generic function

Place a single type variable in angle brackets after the function name. The type variable must appear in the first parameter (directly or as a container element like vector<T>).

```
fn identity<T>(x: T) -> T { x }
```

27.0.2. Calling a generic function

No special syntax at the call site — the compiler infers T from the first argument's type and creates a specialised copy automatically.

```
fn test_identity() {
  assert(identity(42) == 42, "identity integer");
  assert(identity(3.14) == 3.14, "identity float");
  assert(identity("hello") == "hello", "identity text");
  assert(identity(true) == true, "identity bool");
}
```

27.0.3. Multiple parameters of the same type

Additional parameters can also use T. They all share the same concrete type.

```
fn pick_second<T>(gen_a: T, gen_b: T) -> T {
  _x = gen_a;
  gen_b
}
fn test_pick_second() {
  assert(pick_second(1, 99) == 99, "pick second int");
  assert(pick_second("a", "b") == "b", "pick second text");
}
```

27.0.4. Generic functions on vectors

The most common use of generics: write a function that works on any vector. T is inferred from the vector's element type.

```
fn first_element<T>(gen_v: vector<T>) -> T {
  gen_v[0]
}
fn last_element<T>(gen_v: vector<T>) -> T {
  gen_v[gen_v.len() - 1]
}
fn test_vector_generics() {
  ints = [10, 20, 30];
  assert(first_element(ints) == 10, "first int");
  assert(last_element(ints) == 30, "last int");
```

```
  words = ["alpha", "beta", "gamma"];
  assert(first_element(words) == "alpha", "first text");
  assert(last_element(words) == "gamma", "last text");
}
```

27.0.5. Bounded generics with interfaces

By default, you can only assign and return T — no arithmetic, no comparison. To use operators on T, add an interface bound: `\<T: Ordered\>` means T must support `\<`, `\<=`, `\>`, `\>=` comparisons.

Built-in interfaces:

```
Ordered    — comparison operators (<, <=, >, >=)
Equatable  — equality operators (==, !=)
Addable    — addition and subtraction (+, -)
Numeric    — all arithmetic (+, -, *, /, %)
Scalable   — multiplication by integer (* integer)
```

```
fn gen_max<T: Ordered>(gen_x: T, gen_y: T) -> T {
  if gen_x > gen_y { gen_x } else { gen_y }
}
fn gen_min<T: Ordered>(gen_x: T, gen_y: T) -> T {
  if gen_x < gen_y { gen_x } else { gen_y }
}
fn test_bounded() {
  assert(gen_max(3, 7) == 7, "max int");
  assert(gen_max(2.5, 1.5) == 2.5, "max float");
  assert(gen_min(3, 7) == 3, "min int");
  assert(gen_min("apple", "banana") == "apple", "min text");
}
```

27.0.6. Combining bounds

Use + to require multiple interfaces: `\<T: Ordered + Addable\>`.

```
fn clamped_add<T: Ordered + Addable>(gen_a: T, gen_b: T, gen_hi: T) -> T {
  result = gen_a + gen_b;
  if result > gen_hi { gen_hi } else { result }
}
fn test_combined_bounds() {
  assert(clamped_add(3, 4, 10) == 7, "under limit");
  assert(clamped_add(8, 5, 10) == 10, "clamped to limit");
  assert(clamped_add(1.0, 2.0, 2.5) == 2.5, "float clamped");
}
```

27.0.7. Allowed operations on T

Inside a generic function you may only use operations that the bound guarantees. Without any bound: assign, return, store in variables. With Ordered: comparisons. With Addable: + and -.

27.0.8. Disallowed operations

The compiler rejects operations not covered by the bound. `x + y` on unbounded T gives:

```
"generic type T: operator '+' requires a concrete type"
```

`x.field` on T gives:

```
"generic type T: field access requires a concrete type"
```

```
fn main() {
  test_identity();
  test_pick_second();
  test_vector_generics();
  test_bounded();
  test_combined_bounds();
}
```

28. Closures

#warn Dead assignment — 'base' is overwritten before being read @TITLE: Capturing variables from the surrounding scope A closure is a lambda that reads variables from the function scope in which it is written. No explicit capture list is needed — the compiler detects which outer variables the lambda body uses and packages them automatically.

28.0.1. Integer capture

A lambda may read any integer variable in scope. Store the lambda, then call it by name.

28.0.2. Multiple integer captures

A lambda can read more than one outer variable at once. All are captured at the moment the lambda is written.

28.0.3. Text capture

Text values are captured by deep-copy, so they are independent of the original variable after capture.

28.0.4. Capture timing

Loft captures variables at the moment the lambda is written (definition time), not when it is called. If a variable changes after the lambda is written, the lambda still sees the original value.

28.0.5. Cross-scope closures

A function can return a capturing lambda to the caller. The captured values travel with the lambda — no dangling references.

28.0.6. Closures with higher-order functions

A capturing closure stored in a variable can be called directly.

28.0.7. Non-capturing lambdas with higher-order functions

Lambdas that use only their own parameters (no capture) also work fine.

```
fn make_adder(base_val: integer) -> fn(integer) -> integer {
  fn(n: integer) -> integer { base_val + n }
}
```

```
fn main() {
```

28.0.8. Integer capture

```
  offset = 100;
  shift = fn(x: integer) -> integer { x + offset };
  assert(shift(5) == 105, "shift(5): {shift(5)}");
  assert(shift(-3) == 97, "shift(-3): {shift(-3)}");
```

28.0.9. Multiple integer captures

```
  lo = 10;
  hi = 20;
  clamp_val = fn(x: integer) -> integer {
```

```
  if x < lo { lo } else if x > hi { hi } else { x }
};
assert(clamp_val(5) == 10, "clamp below: {clamp_val(5)}");
assert(clamp_val(15) == 15, "clamp mid: {clamp_val(15)}");
assert(clamp_val(25) == 20, "clamp above: {clamp_val(25)}");
```

28.0.10. Text capture

```
greeting = "Hello";
make_msg = fn(name: text) -> text { "{greeting}, {name}!" };
assert(make_msg("World") == "Hello, World!", "greeting capture");
assert(make_msg("Loft") == "Hello, Loft!", "greeting capture 2");
```

28.0.11. Capture timing

Closures capture at definition time. 'base' is 10 when the lambda is written; reassigning 'base' to 20 afterwards does not affect the captured value.

```
base = 10;
add_base = fn(n: integer) -> integer { base + n };
base = 20;
assert(add_base(5) == 15, "sees base=10 at definition time: {add_base(5)}");
```

28.0.12. Cross-scope closures

make_adder returns a lambda that captured base_val from its parameter.

```
add10 = make_adder(10);
add100 = make_adder(100);
assert(add10(5) == 15, "add10(5): {add10(5)}");
assert(add100(5) == 105, "add100(5): {add100(5)}");
```

28.0.13. Closures with higher-order functions

```
factor = 3;
scale = fn(x: integer) -> integer { x * factor };
assert(scale(5) == 15, "scale(5): {scale(5)}");
assert(scale(10) == 30, "scale(10): {scale(10)}");
```

28.0.14. Non-capturing lambdas with higher-order functions

No capture needed here — the lambda uses only its own parameter.

```
nums = [1, 2, 3, 4, 5];
doubled = map(nums, |x| { x * 2 });
assert(doubled[0] == 2, "doubled[0]: {doubled[0]}");
assert(doubled[4] == 10, "doubled[4]: {doubled[4]}");
```

```
evens = filter(nums, |x| { x % 2 == 0 });
assert(evens[0] == 2, "evens[0]: {evens[0]}");
```

```
  assert(evens[1] == 4, "evens[1]: {evens[1]}");
}
```

29. Coroutines

#warn Variable d is never read @TITLE: Lazy sequences with generators and yield A generator function produces values one at a time. Declare the return type as ‘iterator<T>’ and use ‘yield’ to emit each value. The function body is suspended between yields and resumed when the next value is requested. When the body finishes the generator is exhausted. Generators are useful when you want to produce values on demand, stop early with ‘break’, or chain sequences without building intermediate collections.

29.0.1. Declaring a generator function

A function whose return type is ‘iterator<T>’ is a generator. ‘yield value’ emits one item and pauses; the next call resumes from there.

29.0.2. Iterating with a for loop

A ‘for’ loop drives the generator forward automatically. The body runs once per yielded value. Use ‘break’ to stop early without consuming the rest.

29.0.3. Manual advance with next() and exhausted()

Call ‘next(gen)’ yourself when you need fine-grained control. ‘exhausted(gen)’ returns true once there are no more values.

29.0.4. Generators with parameters

A generator function may take any parameters, including text. Parameter values are preserved across yields so the generator can use them throughout its lifetime.

29.0.5. Delegation with yield from

‘yield from sub_gen()’ forwards all values from another generator inline, as if each of its yields appeared at this point directly.

29.0.6. Early termination with break

Stopping a ‘for’ loop before the generator is exhausted is safe. The abandoned generator frame is freed automatically.

```
fn count_up(start: integer, stop: integer) -> iterator<integer> {
  for i in start..stop {
    yield i;
  }
}
```

```
fn squares(n: integer) -> iterator<integer> {
  for i in 1..=n {
    yield i * i;
  }
}
```

Yields the length of ‘s’ twice — shows that a text parameter survives yields.

```
fn len_twice(s: text) -> iterator<integer> {
  yield len(s);
  yield len(s);
}
```

```
fn inner_vals() -> iterator<integer> {
  yield 10;
  yield 20;
}
```

```
fn outer_combined() -> iterator<integer> {
  yield 1;
  yield from inner_vals();
  yield 2;
}
```

```
fn large_range(limit: integer) -> iterator<integer> {
  for i in 1..=limit {
    yield i;
  }
}
```

```
fn main() {
```

29.0.7. Iterating with a for loop

```
  total = 0;
  for n in count_up(1, 6) {
    total += n;
  }
  assert(total == 15, "sum 1..5: {total}");
```

Squares via for loop with index tracking.

```
  idx = 0;
  for s in squares(4) {
    idx += 1;
    if idx == 1 { assert(s == 1, "sq1: {s}"); }
    if idx == 2 { assert(s == 4, "sq2: {s}"); }
    if idx == 3 { assert(s == 9, "sq3: {s}"); }
    if idx == 4 { assert(s == 16, "sq4: {s}"); }
  }
  assert(idx == 4, "squares count: {idx}");
```

29.0.8. Manual advance with next() and exhausted()

```
  gen = count_up(10, 13);
  a = next(gen);
  b = next(gen);
  c = next(gen);
  assert(a == 10 && b == 11 && c == 12, "manual next: {a},{b},{c}");
```

One extra advance past the last element; exhausted() is true after that.

```
  d = next(gen);
  done = exhausted(gen);
  assert(done, "exhausted after last value");
```

29.0.9. Generators with parameters

'len_twice' takes a text parameter and yields its length twice. The text value is preserved across the yield.

```
  lt_total = 0;
  for n in len_twice("hello") {
    lt_total += n;
  }
  assert(lt_total == 10, "len_twice: {lt_total}");
```

29.0.10. Delegation with yield from

```
  yf_total = 0;
  for n in outer_combined() {
    yf_total += n;
  }
  assert(yf_total == 33, "yield from: 1+10+20+2={yf_total}");
```

29.0.11. Early termination with break

Stop after the first 5 values from a 1000-element generator.

```
  limit_total = 0;
  for n in large_range(1000) {
    if n > 5 { break; }
    limit_total += n;
  }
  assert(limit_total == 15, "early break sum 1..5: {limit_total}");
}
```

30. Tuples

A tuple groups a fixed number of values of possibly different types. You do not need to give a tuple a name — use it directly where you need to pass or return multiple values together. Access individual elements with '.0', '.1', etc.

30.0.1. Creating tuples

Write the elements in parentheses, separated by commas. The compiler infers the element types from what you put in. You can add an explicit type annotation when needed.

30.0.2. Accessing elements

Use '.0', '.1', '.2', … to read individual elements.

30.0.3. Modifying elements

Tuple elements can be reassigned like ordinary variables.

30.0.4. Tuples as function parameters

Pass a tuple to a function by value (a copy) or by reference. A value parameter gets its own copy — changes inside the function do not affect the caller. A reference parameter ('&') gives the function a direct link to the caller's tuple so it can modify individual elements in place.

30.0.5. Returning tuples

A function can return a tuple to give back more than one value at once.

30.0.6. Destructuring

Assign a tuple to multiple names in one step using the '(a, b) = expr' form. This is concise when a function returns a tuple and you need both values.

30.0.7. Three or more elements

Tuples can have any number of elements.

```
fn min_max(lo: integer, hi: integer) -> (integer, integer) {
  if lo <= hi {
    (lo, hi)
  } else {
    (hi, lo)
  }
}
```

```
fn swap_ints(pair: &(integer, integer)) {
  tmp = pair.0;
  pair.0 = pair.1;
  pair.1 = tmp;
}
```

```
fn main() {
```

30.0.8. Creating tuples

```
t = (10, 20);
assert(t.0 == 10, "t.0: {t.0}");
assert(t.1 == 20, "t.1: {t.1}");
```

Type annotation

```
w: (integer, integer) = (3, 4);
assert(w.0 + w.1 == 7, "annotated sum: {w.0+w.1}");
```

30.0.9. Modifying elements

```
v = (1, 2);
v.0 = 10;
assert(v.0 + v.1 == 12, "after assign: {v.0}+{v.1}");
```

30.0.10. Tuples as value parameters

```
p = (10, 20);
sum = p.0 + p.1;
assert(sum == 30, "sum: {sum}");
```

30.0.11. Tuples as reference parameters (swap in place)

```
pair = (3, 7);
swap_ints(pair);
assert(pair.0 == 7 && pair.1 == 3, "after swap: ({pair.0},{pair.1})");
```

30.0.12. Returning a tuple from a function

```
bounds = min_max(5, 2);
assert(bounds.0 == 2, "min: {bounds.0}");
assert(bounds.1 == 5, "max: {bounds.1}");
```

Already in order

```
bounds2 = min_max(1, 9);
assert(bounds2.0 == 1, "min2: {bounds2.0}");
assert(bounds2.1 == 9, "max2: {bounds2.1}");
```

30.0.13. Destructuring

```
(lo, hi) = min_max(8, 3);
assert(lo == 3, "destructured lo: {lo}");
assert(hi == 8, "destructured hi: {hi}");
```

30.0.14. Three or more elements

```
  u = (1, 2, 3);
  assert(u.0 + u.1 + u.2 == 6, "triple sum: {u.0}+{u.1}+{u.2}");
```

```
  triple = (10, 20, 30);
  triple.1 = 99;
  assert(triple.0 == 10 && triple.1 == 99 && triple.2 == 30,
    "triple modified: ({triple.0},{triple.1},{triple.2})");
}
```

31. Match

The `match` expression lets you compare a value against a series of patterns and run different code for each case. It is similar to `switch` in other languages but more powerful: patterns can destructure structs, bind fields to local variables, and include `if` guards. Match is an expression — it returns a value, so you can assign the result or use it directly inside a larger expression.

31.0.1. Simple enum matching

The most common use: branch on an enum variant. Every variant must be covered or a _ wildcard must appear — the compiler checks exhaustiveness.

```
enum Direction { North, South, East, West }
```

```
fn direction_name(d: Direction) -> text {
  match d {
    North => "north",
    South => "south",
    East  => "east",
    West  => "west",
  }
}
```

```
fn main() {
  assert(direction_name(North) == "north", "simple enum match");
  assert(direction_name(West) == "west", "west");
```

31.0.2. Match as expression

Because match returns a value you can use it in assignments and arguments.

```
  score = match Direction.East {
    North | South => 10,
    East | West   => 20,
  };
  assert(score == 20, "match expression: {score}");
```

31.0.3. Struct-enum destructuring

When an enum has variants with fields (a struct-enum) you can bind the fields to variables in the match arm.

```
Circle { radius } — binds the 'radius' field to a local variable
Rect { w, h }     — binds both 'w' and 'h'
```

The variable names must match the field names exactly.

31.0.4. Guards

Add `if condition` after a pattern to restrict when the arm matches. The guard can reference variables bound by the pattern.

```
v = 42;
label = match v {
  0      => "zero",
  _ if v < 0 => "negative",
  _ if v > 100 => "large",
  _      => "normal",
};
assert(label == "normal", "guard: {label}");
```

31.0.5. Wildcard _

The underscore _ matches anything. Put it last as a catch-all. Without it, the compiler will reject the match if any value could fall through without matching.

```
x = 7;
kind = match x {
  1 => "one",
  2 => "two",
  _ => "other",
};
assert(kind == "other", "wildcard: {kind}");
```

31.0.6. Integer matching

Match works on integers — each arm is compared with ==.

```
name = match 3 {
  1 => "one",
  2 => "two",
  3 => "three",
  _ => "many",
};
assert(name == "three", "integer match: {name}");
```

31.0.7. Text matching

Match also works on text values.

```
greeting = "hi";
reply = match greeting {
  "hello" => "formal",
  "hi"    => "casual",
  _       => "unknown",
};
assert(reply == "casual", "text match: {reply}");
```

31.0.8. Multiple patterns with |

Combine patterns with | to share the same arm body.

```
  d = Direction.South;
  axis = match d {
    North | South => "vertical",
    East | West   => "horizontal",
  };
  assert(axis == "vertical", "multi-pattern: {axis}");
```

31.0.9. Nested match

Match can appear inside other expressions, including other match arms.

```
  n = 15;
  result = match n % 2 == 0 {
    true  => "even",
    false => match n % 3 == 0 {
      true  => "odd multiple of 3",
      false => "odd",
    },
  };
  assert(result == "odd multiple of 3", "nested: {result}");
}
```

32. Formatting

Loft strings can embed expressions inside `{...}` braces. A colon after the expression introduces a format specifier that controls width, alignment, precision, number base, and output style.

32.0.1. Basic interpolation

Any expression inside `{...}` is evaluated and converted to text.

```
fn main() {
  name = "world";
  assert("hello {name}" == "hello world", "basic interpolation");
  assert("1 + 2 = {1 + 2}" == "1 + 2 = 3", "expression in braces");
```

32.0.2. Width and alignment

A number after `:` sets the minimum width. The value is padded with spaces to fill the width.

```
{val:6}   — right-aligned (default for numbers)
{val:>6}  — right-aligned (explicit)
{val:<6}  — left-aligned
{val:^6}  — centered
```

For text, the default alignment is left. For numbers, right.

```
  assert("{42:6}" == "    42", "default number align is right");
  assert("{42:>6}" == "    42", "explicit right-align");
  assert("{42:<6}" == "42    ", "left-align number");
  assert("{42:^6}" == "  42  ", "center-align number");
  s = "hi";
  assert("{s:6}" == "hi    ", "default text align is left");
  assert("{s:>6}" == "    hi", "right-align text");
  assert("{s:^6}" == "  hi  ", "center-align text");
```

32.0.3. Zero padding

Prefix the width with `0` to pad with zeros instead of spaces.

```
  assert("{7:03}" == "007", "zero-padded 3 digits");
  assert("{42:05}" == "00042", "zero-padded 5 digits");
  assert("{-1:04}" == "-001", "zero-padded negative: sign before zeros");
```

32.0.4. Signed format

`+` forces a sign on positive numbers.

```
  assert("{42:+}" == "+42", "explicit positive sign");
  assert("{-42:+}" == "-42", "negative sign always shown");
  assert("{0:+}" == "+0", "sign on zero");
```

32.0.5. Hexadecimal, binary, octal

`:x` for lowercase hex, `:\#x` for hex with `0x` prefix. `:b` for binary, `:o` for octal.

```
assert("{255:x}" == "ff", "hex lowercase");
assert("{255:#x}" == "0xff", "hex with prefix");
assert("{10:b}" == "1010", "binary");
assert("{8:o}" == "10", "octal");
```

32.0.6. Float precision

`.N` after the colon limits decimal places.

```
assert("{3.125:.1}" == "3.1", "1 decimal place");
assert("{3.125:.2}" == "3.12", "2 decimal places");
assert("{3.125:.0}" == "3", "0 decimal places");
assert("{0.0:.3}" == "0.000", "trailing zeros");
```

32.0.7. Width + precision

Combine width and precision: `{val:W.P}` where W is total width and P is decimal places.

```
assert("{3.125:8.2}" == "    3.12", "width 8 precision 2");
assert("{-3.125:8.2}" == "   -3.12", "negative width+precision");
```

32.0.8. JSON format

`:j` serialises a struct or value as JSON. `:j` serialises a struct as JSON. Use it on struct values, not primitives. See the JSON documentation page for full details.

32.0.9. Vector format

Vectors are formatted as `\[a,b,c\]` by default. A format specifier inside a `for` loop applies to each element.

```
v = [1, 2, 3];
assert("{v}" == "[1,2,3]", "default vector format");
assert("{for fmt_n in 1..4 {fmt_n * 10}:04}" == "[0010,0020,0030]", "formatted
vector elements");
```

32.0.10. Long and single types

Long integers and single-precision floats use the same format specifiers.

```
n = 1000000l;
assert("{n}" == "1000000", "long default");
assert("{n:>10}" == "   1000000", "long right-aligned");
f = 1.5f;
assert("{f}" == "1.5", "single default");
```

32.0.11. Character format

```
c = 'A';
assert("{c}" == "A", "character default");
```

32.0.12. Boolean format

```
  assert("{true}" == "true", "boolean true");
  assert("{false}" == "false", "boolean false");
}
```

33. Standard Library

33.1. Types

```
pub type boolean size(1)
```

Primitive types built into lav. True or false value.

```
pub type integer size(4)
```

32-bit signed integer.

```
pub type long size(8)
```

64-bit signed integer. Use when values exceed 2 billion.

```
pub type single size(4)
```

32-bit floating-point. Good for graphics and performance-sensitive math.

```
pub type float size(8)
```

64-bit floating-point. Use when precision matters.

```
pub type text size(4)
```

UTF-8 string.

```
pub type character size(4)
```

A single Unicode code point.

```
pub type u8 = integer limit(0, 255) size(1)
```

Integer subtypes for compact storage in struct fields. Behave as integer in expressions. 0 – 255, 1 byte.

```
pub type i8 = integer limit(-128, 127) size(1)
```

–128 – 127, 1 byte.

```
pub type u16 = integer limit(0, 65535) size(2)
```

0 – 65535, 2 bytes.

```
pub type i16 = integer limit(-32768, 32767) size(2)
```

–32768 – 32767, 2 bytes.

```
pub type i32 = integer size(4)
```

Full 32-bit integer range, 4 bytes.

33.2. Interfaces

Standard interfaces for bounded generic functions. A type satisfies an interface by defining the required operator or method.

```
pub interface Ordered
```

Types that support the \< comparison operator. Satisfied by integer, long, single, float, text, and any user type defining OpLt.

```
pub interface Equatable
```

Types that support the == equality operator. Satisfied by integer, long, single, float, text, boolean, and user types defining OpEq.

```
pub interface Addable
```

Types that support the + addition operator, returning the same type. Satisfied by integer, long, single, float, and user types defining OpAdd.

```
pub interface Numeric
```

Types that support * and - (unary negation). Separate from `Addable` to allow fine-grained bounds without stub-name collisions. Satisfied by integer, long, single, float, and user types defining OpMul and OpMin.

```
pub interface Scalable
```

Types that support integer scaling via a `scale` method. Uses a method (not `op \*`) to avoid stub-name collision with `Numeric`. User types satisfy Scalable by defining `fn scale(self: T, factor: integer) -\> integer`.

```
pub interface Printable
```

Types that can be converted to text via a `to\_text` method. User types satisfy Printable by defining `fn to\_text(self: T) -\> text`.

33.3. Math

Functions for numeric computation. All trigonometric functions work in radians. Both single and float variants exist for every function — choose single for speed, float for precision. Integer operations

```
pub fn abs(both: integer) -> integer
```

Absolute value. Removes the sign from a negative integer.

```
pub fn abs(both: long) -> long
```

Absolute value for long integers.

```
pub fn abs(both: single) -> single
```

Absolute value for single-precision floats.

```
pub fn cos(both: single) -> single
```

Cosine. Use for circular motion: x = r * cos(angle).

```
pub fn sin(both: single) -> single
```

```
pub fn tan(both: single) -> single
```

```
pub fn acos(both: single) -> single
```

```
pub fn asin(both: single) -> single
```

```
pub fn atan(both: single) -> single
```

```
pub fn ceil(both: single) -> single
```

```
pub fn floor(both: single) -> single
```

```
pub fn round(both: single) -> single
```

```
pub fn sqrt(both: single) -> single
```

```
pub fn atan2(both: single, v2: single) -> single
```

Arc tangent of y/x, preserving the correct quadrant. Use instead of atan when you have separate x/y components.

```
pub fn log(both: single, v2: single) -> single
```

```
pub fn pow(both: single, v2: single) -> single
```

Raises base to the power exp. Use for exponential growth curves and scaling.

```
pub fn abs(both: float) -> float
```

Absolute value for double-precision floats.

```
pub PI = OpMathPiFloat()
```

The ratio of a circle's circumference to its diameter (3.14159...).

```
pub E = OpMathEFloat()
```

Euler's number, the base of natural logarithms (2.71828...).

```
pub fn cos(both: float) -> float
```

Cosine. Use for circular motion: x = r * cos(angle).

```
pub fn sin(both: float) -> float
```

```
pub fn tan(both: float) -> float
```

```
pub fn acos(both: float) -> float
```

```
pub fn asin(both: float) -> float
```

```
pub fn atan(both: float) -> float
```

```
pub fn ceil(both: float) -> float
```

```
pub fn floor(both: float) -> float
```

```
pub fn round(both: float) -> float
```

```
pub fn sqrt(both: float) -> float
```

```
pub fn atan2(both: float, v2: float) -> float
```

Arc tangent of y/x, preserving the correct quadrant. Use instead of atan when you have separate x/y components.

```
pub fn log(both: float, v2: float) -> float
```

```
pub fn pow(both: float, v2: float) -> float
```

Raises base to the power exp. Use for exponential growth curves and scaling.

33.4. exp / ln / log2 / log10

Raises E (2.71828...) to the power v. Use for exponential growth models.

```
pub fn exp(both: single) -> single
```

```
pub fn exp(both: float) -> float
```

```
pub fn ln(both: single) -> single
```

```
pub fn ln(both: float) -> float
```

```
pub fn log2(both: single) -> single
```

```
pub fn log2(both: float) -> float
```

```
pub fn log10(both: single) -> single
```

```
pub fn log10(both: float) -> float
```

33.5. min / max / clamp

```
pub fn min(both: integer, b: integer) -> integer
```

Returns null if either argument is null (consistent with all binary arithmetic operations). Smallest of two integer values.

```
pub fn max(both: integer, b: integer) -> integer
```

```
pub fn clamp(both: integer, lo: integer, hi: integer) -> integer
```

```
pub fn min(both: long, b: long) -> long
```

```
pub fn max(both: long, b: long) -> long
```

```
pub fn clamp(both: long, lo: long, hi: long) -> long
```

```
pub fn min(both: single, b: single) -> single
```

```
pub fn max(both: single, b: single) -> single
```

```
pub fn clamp(both: single, lo: single, hi: single) -> single
```

```
pub fn min(both: float, b: float) -> float
```

```
pub fn max(both: float, b: float) -> float
```

```
pub fn clamp(both: float, lo: float, hi: float) -> float
```

33.6. Text

Functions for working with text (UTF-8 strings) and character values. Read the value of a variable and put a reference to it on the stack

```
pub fn len(both: text) -> integer
```

Number of bytes in the text. Use for bounds checks and iteration limits.

```
pub fn size(both: text) -> integer
```

Number of Unicode code points (characters) in the text.

```
pub fn len(both: character) -> integer
```

Byte length of the character's UTF-8 encoding (1–4).

Splits self on every occurrence of separator and returns the parts as a vector. Use to parse CSV lines or space-separated tokens.

```
pub fn split(self: text, separator: character) -> vector < text >
```

```
pub fn join(self: vector<text>, jn_sep: text) -> text
```

```
pub fn starts_with(self: text, value: text) -> boolean
```

Functions for searching, transforming, and classifying text and character values. Character classification functions return true only if every character in the text satisfies the condition. The single-character variants test one code point. Returns true if self begins with value. Use for prefix matching (e.g., protocol detection).

```
pub fn ends_with(self: text, value: text) -> boolean
```

Returns true if self ends with value. Use for suffix matching (e.g., file extension checks).

```
pub fn trim(both: text) -> text[both]
```

Removes leading and trailing whitespace. Use when processing user input or file content.

```
pub fn trim_start(self: text) -> text[self]
```

Removes leading whitespace only.

```
pub fn trim_end(self: text) -> text[self]
```

Removes trailing whitespace only.

```
pub fn find(self: text, value: text) -> integer
```

Returns the byte index of the first occurrence of value, or null if not found. Use to locate substrings before slicing.

```
pub fn rfind(self: text, value: text) -> integer
```

Returns the byte index of the last occurrence of value, or null if not found. Use to find file extensions or the last path separator.

```
pub fn contains(self: text, value: text) -> boolean
```

Returns true if value appears anywhere in self. Simpler than find when you only need a yes/no answer.

```
pub fn replace(self: text, value: text, with: text) -> text
```

Returns a copy of self with every occurrence of value replaced by with.

```
pub fn to_lowercase(self: text) -> text
```

Returns a lowercase copy. Use for case-insensitive comparisons.

```
pub fn to_uppercase(self: text) -> text
```

Returns an uppercase copy.

```
pub fn is_lowercase(self: text) -> boolean
```

True if all characters are lowercase letters.

```
pub fn is_lowercase(self: character) -> boolean
```

True if the character is a lowercase letter.

```
pub fn is_uppercase(self: text) -> boolean
```

True if all characters are uppercase letters.

```
pub fn is_uppercase(self: character) -> boolean
```

True if the character is an uppercase letter.

```
pub fn is_numeric(self: text) -> boolean
```

True if all characters are numeric digits (Unicode numeric, not just ASCII 0–9).

```
pub fn is_numeric(self: character) -> boolean
```

True if the character is a numeric digit.

```
pub fn is_alphanumeric(self: text) -> boolean
```

True if all characters are letters or digits. Use to validate identifiers or tokens.

```
pub fn is_alphanumeric(self: character) -> boolean
```

True if the character is a letter or digit.

```
pub fn is_alphabetic(self: text) -> boolean
```

True if all characters are alphabetic.

```
pub fn is_alphabetic(self: character) -> boolean
```

True if the character is alphabetic.

```
pub fn is_whitespace(self: text) -> boolean
```

True if all characters are whitespace. Use to detect blank lines.

```
pub fn is_whitespace(self: character) -> boolean
```

True if the character is whitespace.

```
pub fn is_control(self: text) -> boolean
```

True if all characters are control characters.

```
pub fn is_control(self: character) -> boolean
```

True if the character is a control character.

```
pub fn join(parts: vector < text >, sep: text) -> text
```

Joins parts with sep between each consecutive pair. Returns “” for an empty vector. Use to build comma-separated lists, path segments, or any delimited output.

33.7. Collections

Operations on vector<T> — the primary ordered collection type. Vectors are grown by appending with += and elements are accessed by index. All structures are passed by reference instead of by value

```
pub fn len(both: vector) -> integer
```

Number of elements in the vector. Use in loop bounds: for i in 0..v.len().

```
pub fn clear(both: vector)
```

Remove all elements from the vector, setting its length to 0.

33.8. Output and Diagnostics

```
pub fn assert(test: boolean, message: text, file: text, line: integer)
```

Panics with message if test is false. Use to verify invariants during development. In production mode (–production), logs an error instead of aborting. The file and line are injected by the compiler; do not pass them manually.

```
pub fn panic(message: text, file: text, line: integer)
```

Immediately terminates execution with message. Use for unrecoverable error states. In production mode (–production), logs a fatal entry instead of aborting. The file and line are injected by the compiler; do not pass them manually.

```
pub fn log_info(message: text, file: text, line: integer)
```

Write a structured log record at the chosen severity. The file and line are injected by the compiler; do not pass them manually.

```
pub fn log_warn(message: text, file: text, line: integer)
```

```
pub fn log_error(message: text, file: text, line: integer)
```

```
pub fn log_fatal(message: text, file: text, line: integer)
```

```
pub fn print(v1: text)
```

Writes v to standard output without a newline. Use for progress output and building up a line incrementally.

```
pub fn println(v1: text)
```

33.9. Vector aggregates

Internal helpers for reduce-based aggregates.

```
pub fn sum_of(v: vector<integer>) -> integer
```

Sum of all integer elements. Returns 0 for an empty vector.

```
pub fn min_of<T: Ordered>(v: vector<T>) -> T
```

Smallest element in a non-empty vector. Works on any type satisfying Ordered (op <).

```
pub fn max_of<T: Ordered>(v: vector<T>) -> T
```

Largest element in a non-empty vector. Works on any type satisfying Ordered (op <).

```
pub fn sum<T: Addable>(v: vector<T>, init: T) -> T
```

Sum of vector elements with caller-supplied identity. Works on any Addable type. Example: sum([10, 20, 12], 0) == 42

33.10. Field iteration support types

```
pub enum FieldValue {
  FvBool   { v: boolean },
  FvInt    { v: integer },
  FvLong   { v: long },
  FvFloat  { v: float },
  FvSingle { v: single },
  FvChar   { v: character },
  FvText   { v: text }
}
```

Wraps a field value in a type-discriminated enum for compile-time field iteration.

```
pub struct StructField {
  name:  text not null,
  value: FieldValue,
}
```

Represents a single struct field during compile-time iteration.

33.11. File System

```
pub struct EnvVariable {
  name: text,
  value: text
}
```

Types and functions for reading and writing files. A File value is obtained via file() and carries the path, format, and an internal reference. An environment variable as a name/value pair.

```
pub enum Format {
  TextFile,
  LittleEndian,
  BigEndian,
  Directory,
  NotExists
}
```

```
pub enum FileResult {
  Ok,
  NotFound,
  PermissionDenied,
  IsDirectory,
  NotDirectory,
  Other
}
```

```
pub fn ok(self: FileResult) -> boolean
```

```
pub struct File {
  path: text,

  size: long,

  format: Format,

  ref: i32,

  current: long,

  next: long
}
```

```
pub fn content(self: File) -> text
```

```
pub fn lines(self: File) -> vector < text >
```

Returns the platform path separator character: '\' on Windows, '/' elsewhere. Detected once at startup from the runtime filesystem.

```
pub fn path_sep() -> character
```

```
pub fn file(path: text) -> File
```

```
pub fn exists(path: text) -> boolean
```

```
pub fn exists(both: File) -> boolean
```

Method form: f = file("path"); if f.exists() { ... } Also callable as exists(file_obj) via the 'both' parameter name.

```
pub fn delete(path: text) -> FileResult
```

```
pub fn move(from: text, to: text) -> FileResult
```

```
pub fn mkdir(path: text) -> FileResult
```

```
pub fn mkdir_all(path: text) -> FileResult
```

```
pub fn set_file_size(self: File, size: long) -> FileResult
```

```
pub fn files(self: File) -> vector < File >
```

```
pub fn write(self: File, v: text)
```

33.12. Environment

Returns all environment variables as a vector of EnvVariable records (fields: name, value). Use to inspect or forward the full environment.

```
pub fn env_variables() -> vector < EnvVariable >
```

```
pub fn env_variable(name: text) -> text
```

Returns the value of the environment variable name, or null if it is not set. Use to read configuration from the shell environment.

Functions for interacting with the host operating system. Returns the script-level arguments passed after the script path. Does not include the loft binary name or loft CLI flags.

```
pub fn arguments() -> vector < text >
```

```
pub fn directory(v:  & text = "") -> text
```

Returns the current working directory, optionally with v appended as a subpath. Use to construct absolute paths relative to where the program was launched.

```
pub fn user_directory(v:  & text = "") -> text
```

Returns the current user's home directory, optionally with v appended. Use for storing user-specific data or configuration.

```
pub fn program_directory(v:  & text = "") -> text
```

Returns the directory containing the running executable, optionally with v appended. Use to locate assets bundled alongside the program.

```
pub fn source_dir() -> text
```

Returns the directory containing the main source file being executed. Use to locate data files relative to the script, regardless of working directory.

33.13. Time

```
pub fn now() -> long
```

Returns the current wall-clock time as milliseconds since the Unix epoch (1970-01-01T00:00:00 UTC). Use for timestamps and logging.

```
pub fn ticks() -> long
```

Returns microseconds elapsed since program start (monotonic clock). Unaffected by system clock adjustments. Use for benchmarks and frame timing.

34. Roadmap

Loft is under active development. Everything documented on the language pages works today. This page describes where the project is headed.

The project goal is **browser games that anyone can play via a shared link**. Native OpenGL is supported for desktop enthusiasts. Server and multiplayer features come after the single-player browser experience works.

34.0.1. Current release — 0.8.4: Awesome Brick Buster

0.8.4 turns Brick Buster from a tech demo into a game someone would actually want to share with a friend, and makes sharing trivial via single-file HTML export.

34.0.1.1. Game polish

- **Hand-designed levels** — five themed layouts dispatched by `level_brick(lv, r, c)`, with a procedural fallback for deeper runs.
- **Cel-shaded sprite atlas** — outlined icons, a round ball with velocity-directional squash, hearts for lives, balloon projectiles, fireball after-images.
- **Roman-numeral level display** and **persistent high score** (`.loft/brickbuster_score.txt`).
- **Chiptune soundtrack** — three early-Capcom-style tracks (Heroic / Determined / Calm) that play once per level with silences between, plus brick/paddle/wall/pickup/life sound effects.
- **Screen shake** on brick breaks and life lost; pause menu; title and game-over screens.

34.0.1.2. Single-file HTML export

Compile any loft program to a self-contained `.html` file that runs in any browser — no install, no server.

```
$ loft --html app.loft
```

The game's WebAssembly, textures, and audio are all embedded. Host the file anywhere, send someone the URL, they play.

34.0.1.3. Infrastructure

- **Audio** — `rodio`-backed playback on desktop; WebAudio bridge in the browser.
- **Store-leak fixes** — per-frame idiomatic struct APIs now work without workarounds (P122 + P117–P131 resolved).
- **CLI hardening** — script-level arguments, file-scope constants, and release-mode coroutine-iterator fixes (P126, P128, P131, P132).
- **Bytecode cache** — `.loftc` files are invalidated on interpreter rebuild via the embedded git commit hash.

34.0.2. Next — 0.8.5: Working Moros editor

The Moros hex RPG scene editor runs end-to-end in the browser: load a map, paint hexes, place walls and items, see a live 3D preview, export to GLB. Web only — multiplayer comes in 1.0.0.

- **Map model** — `Hex`, `Chunk`, `Map` with material/wall/item palettes and spawn/NPC routines, serialised as JSON.
- **2D scene canvas** — hex coordinate math, pan/zoom/hit-test, layered rendering.
- **Editor shell** — toolbar, palettes, Paint / Height / Wall / Item tools, undo/redo, localStorage autosave.

- **3D renderer** — hex surface geometry (flat and slope), wall slabs, stairs, camera orbit, hex picking, GLB export.
- **Live 3D preview** alongside the 2D editor, powered by the loft graphics library compiled to WASM.

34.0.3. 0.9.0 — Fully working loft language

The language itself becomes feature-complete, well-documented, and tooling-friendly. No "appears fixed but unverified" bugs. Anyone can write loft code in their preferred editor with syntax highlighting, decent error messages, and a REPL for experimentation.

34.0.3.1. Language polish

- **Error recovery** — the parser continues after a token-level failure and reports multiple errors in a single pass.
- **REPL** — running `loft` with no arguments starts an interactive session; definitions persist across lines.
- **Developer warnings** — Clippy-inspired lints for common mistakes.
- **AOT library compilation** — automatically compile libraries to native shared libs for faster startup.
- **Stdlib hygiene** — name-clash warnings and a `std::` prefix for shadowed builtins; library enums in `match` arms without qualification.

34.0.3.2. Compilation cache

Bytecode cache (`.loftc`) and the shared constant store are already implemented. Remaining work — mmap-based native cache loading, serialised `Data` definitions, pre-compiled stdlib for WASM — lands in 0.9.0.

34.0.3.3. Developer experience

- TextMate grammar for `.loft` syntax highlighting.
- VS Code extension with snippets and a run task.
- Quick-start `examples/` directory.
- CI covering package tests and native codegen.

34.0.3.4. Packaging and FFI

- **Lock file** (`loft.lock`) for reproducible builds.
- **Generic FFI marshaller** — zero-boilerplate native functions from a `#native` signature; generic `cdylib` loader scans exports into a hash map.

34.0.4. 1.0.0 — Totally sure everything works

The stability contract. Anyone can write, run, and share a program — terminal or browser — and trust that it will keep working.

34.0.4.1. Web IDE

A browser-based IDE running the full loft interpreter as WebAssembly — no installation, no server.

- Editor shell with CodeMirror 6 and a loft grammar.
- Symbol navigation (go-to-definition, find-usages).
- Multi-file projects stored locally (IndexedDB).
- Documentation and examples browser built in.
- Export/import ZIP; PWA offline mode.

34.0.4.2. Scene scripting

The Moros scene editor gains a loft scripting panel with in-browser compile and hot reload. Scene scripts are sandboxed, travel with the scene JSON, and can be shared alongside the map.

34.0.4.3. Multiplayer

- **Server library** — plain HTTP routing, HTTPS with static certs or automatic ACME, WebSockets, JWT / session / API-key auth, CORS, rate limiting, static file serving.
- **Game loop primitives** — WebSocket polling, broadcast, connection registry.
- **Game client library** — `GameEnvelope` protocol, lobby + matchmaking, fixed-timestep loop, client-side prediction and reconciliation, WASM script loading with Ed25519 verification.
- **Moros multiplayer** — DM and players share a live scene hosted on a loft server.

34.0.4.4. Stability contract

Version 1.0 means: any program that works on 1.0 will compile and run identically on all future 1.x releases. The language syntax, type system, standard library, and command-line flags are frozen. Until then, breaking changes are possible between minor versions.

Tagging 1.0 requires every stability gate to be satisfied without shortcuts: zero open high-severity bugs, valgrind-clean debug builds of the full Brick Buster and tuple tests, green CI on Linux, macOS Intel, macOS ARM, and Windows, pre-built binaries on the GitHub release, and the reference + PDF linked from the release page.

34.0.5. 1.1 and beyond

Post-1.0 items include route decorators (`@get` / `@post` / `@ws`), WASM Web Worker pools for browser-side `par()`, iterator protocol (`for msg in ws`), interface factory methods, lazy work-variable initialisation, stack raw-pointer cache, spatial index operations, and additional native codegen optimisations.

34.0.6. Following progress

Development is tracked in the GitHub repository. The full internal roadmap with effort estimates, dependencies, and design pointers is in the repository's `doc/claude/ROADMAP.md`.